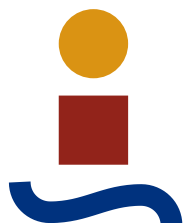


Some Trends in Nuclear Reaction Theory for basic science and applications

Gregory Potel Aguilar
Shanghai, July 15, 2025



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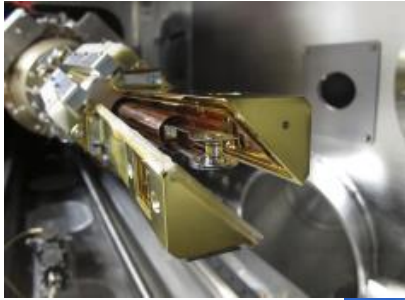
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Part 1

Introduction and Motivation

Nuclear reactions: **why do we care?** Nuclear reactions for applications

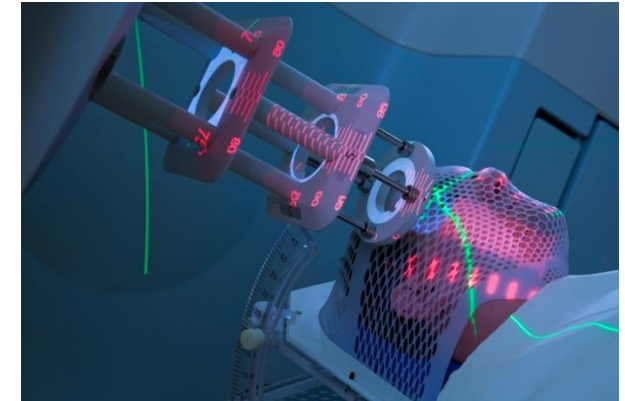
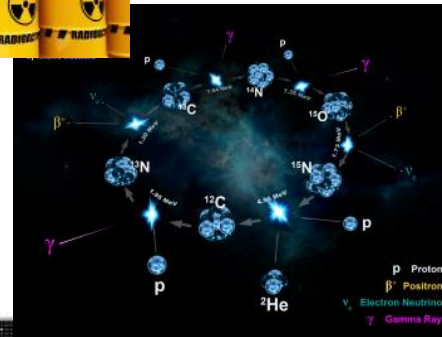
inertial confinement fusion



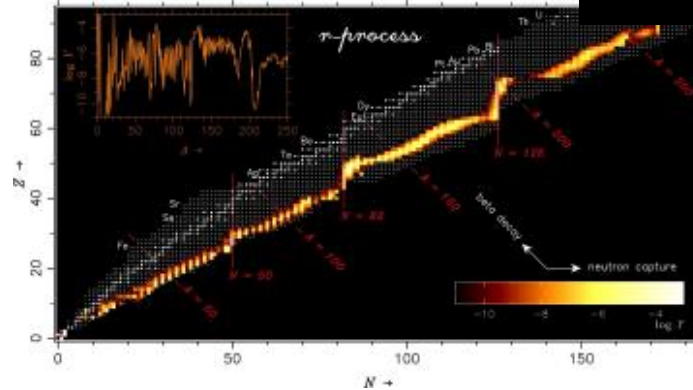
waste management



nuclear reactors



nuclear medicine

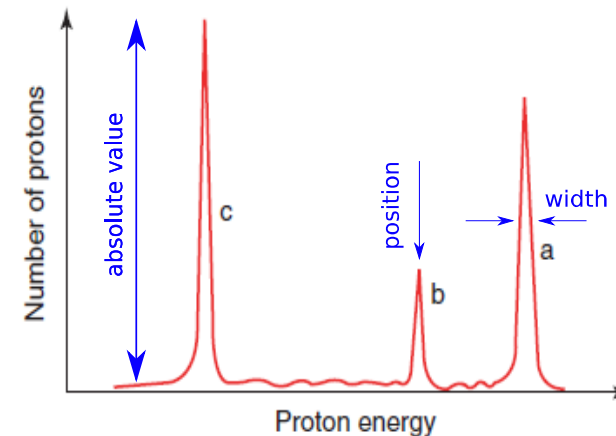
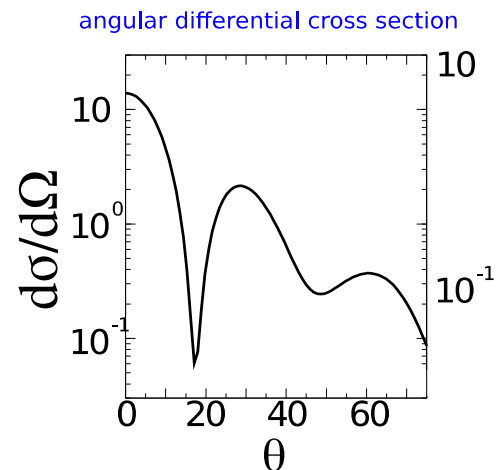
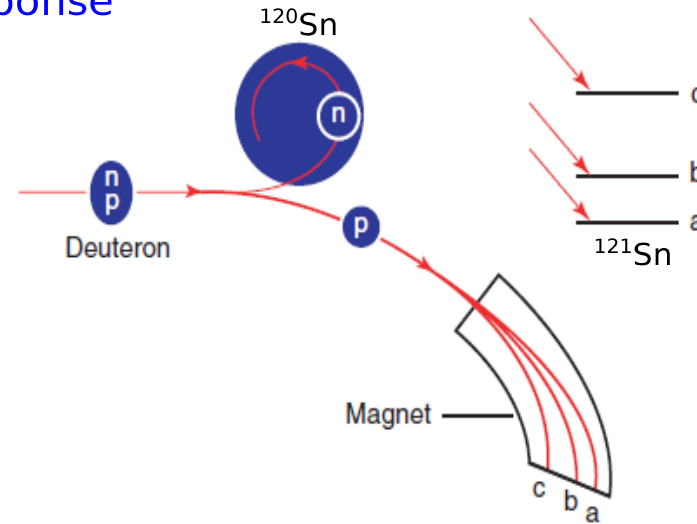


reactions of
astrophysical
interest

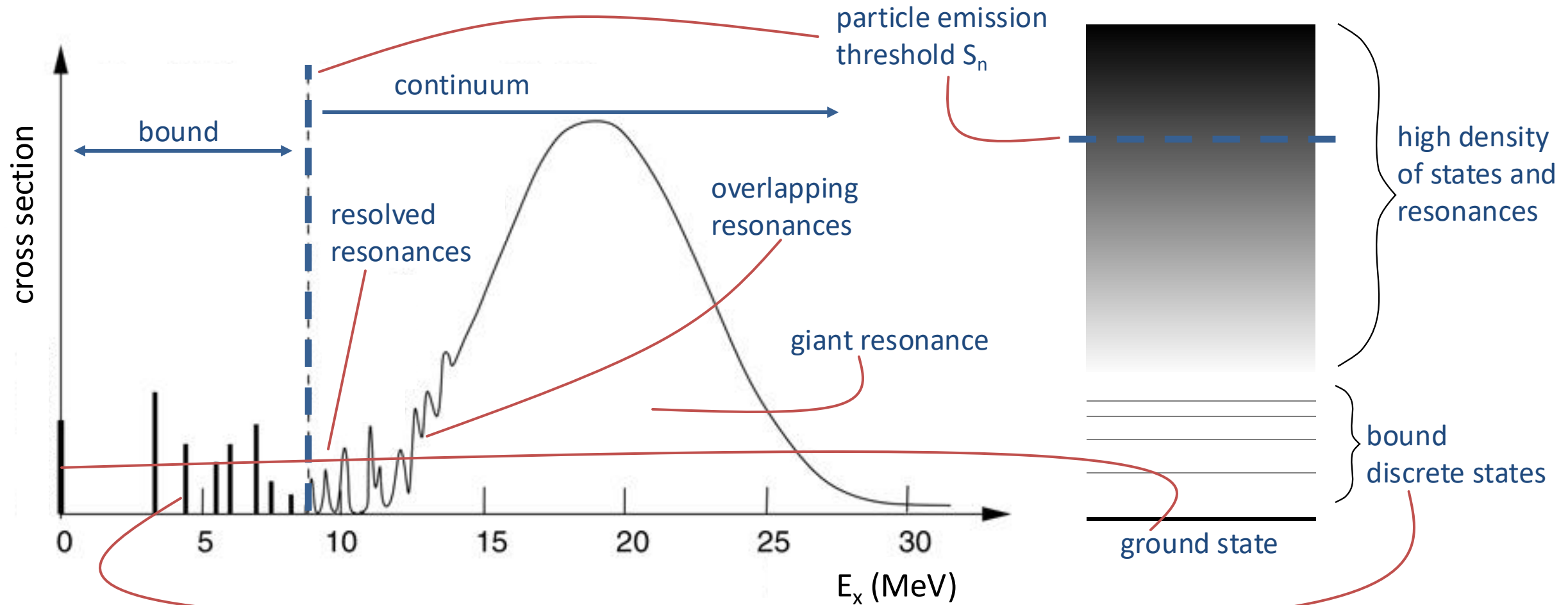
r-process

Nuclear reactions: **why do we care?** Nuclear reactions as an experimental tool

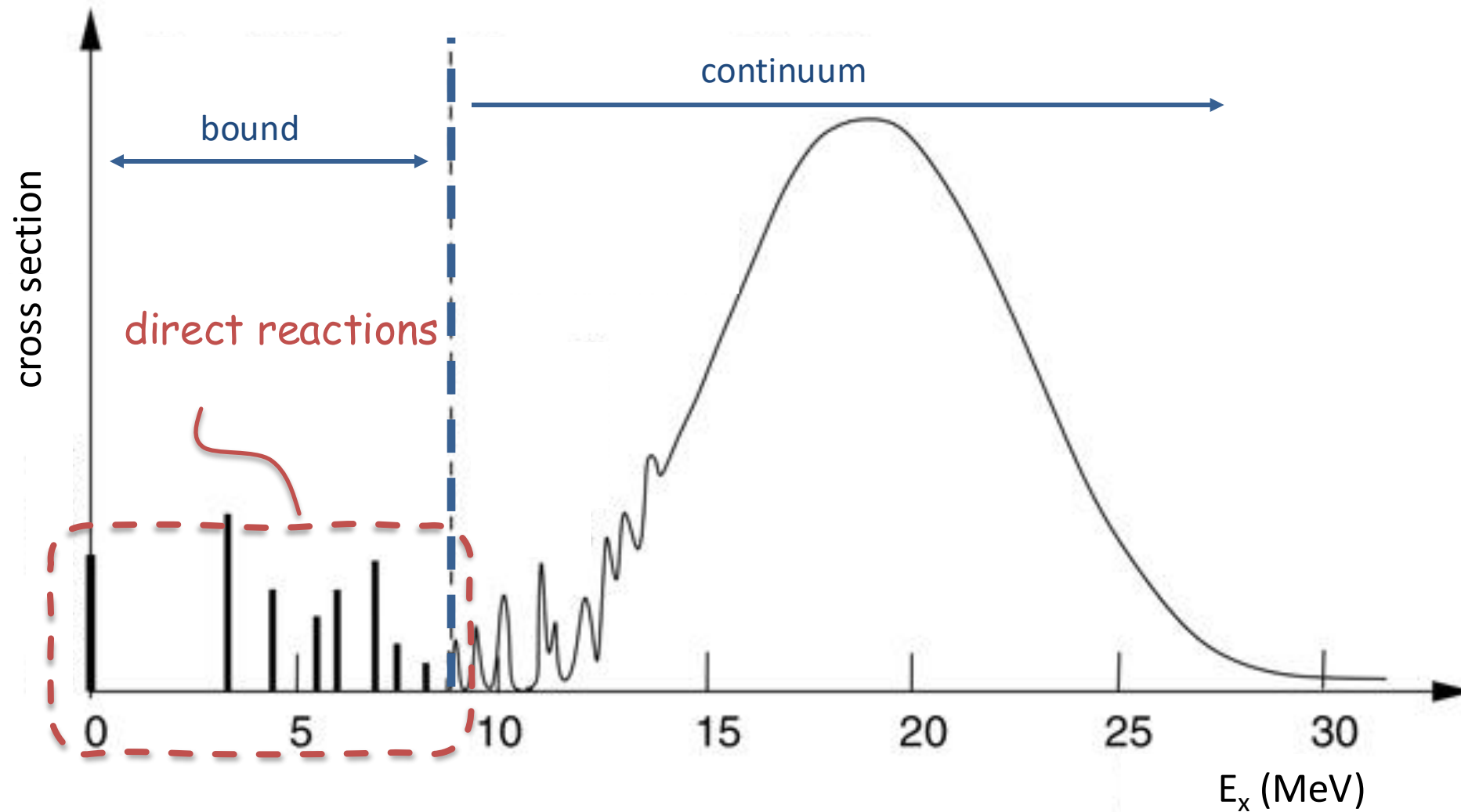
- transfer reactions probe **nuclear response** to the **addition of a nucleon**
- a variety of **observables** provide rich information about **nuclear structure**:
 - **angular differential cross section**
 - **absolute value**
 - **position**
 - **width** (when in the continuum)



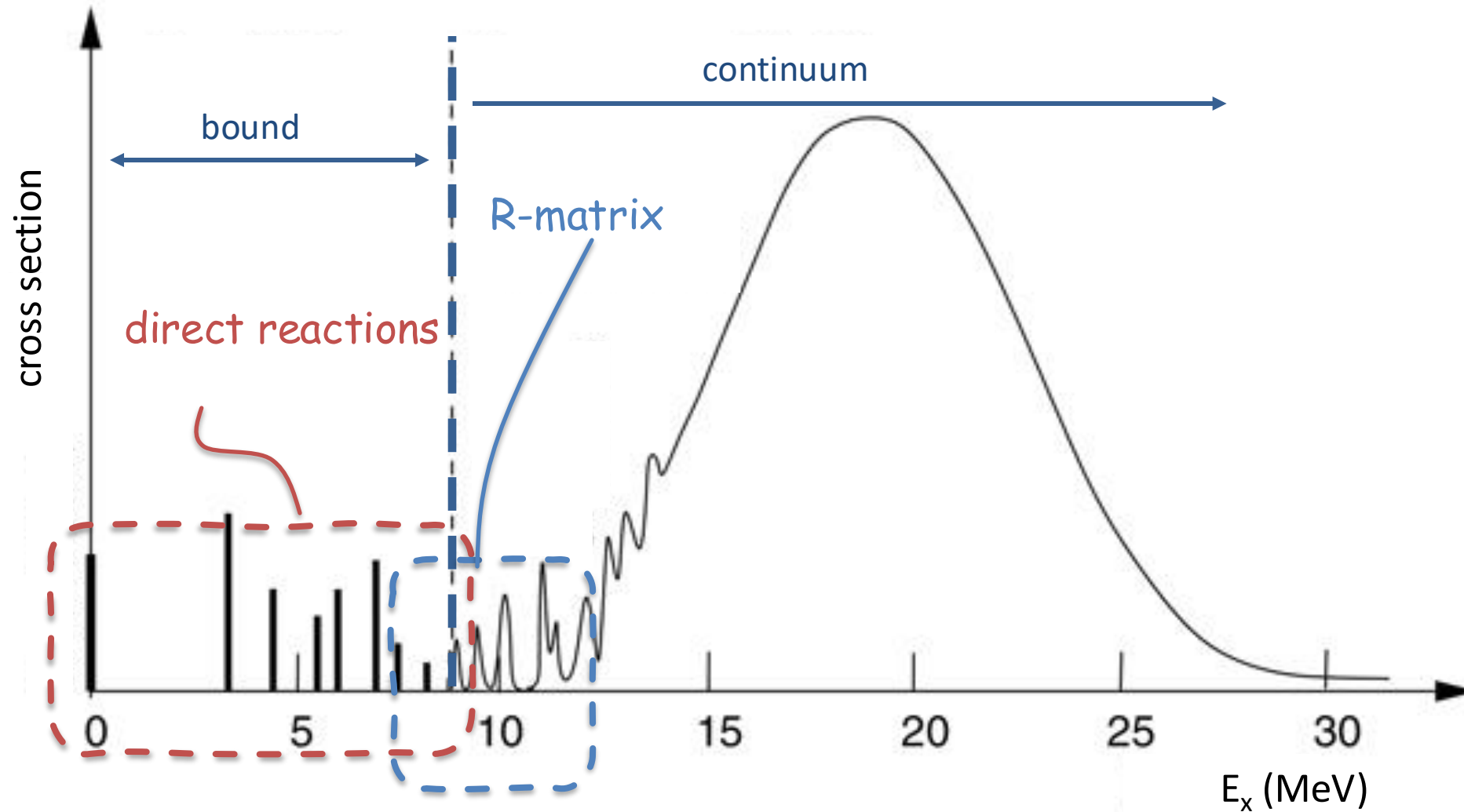
Features of nuclear spectra probed by nuclear reactions



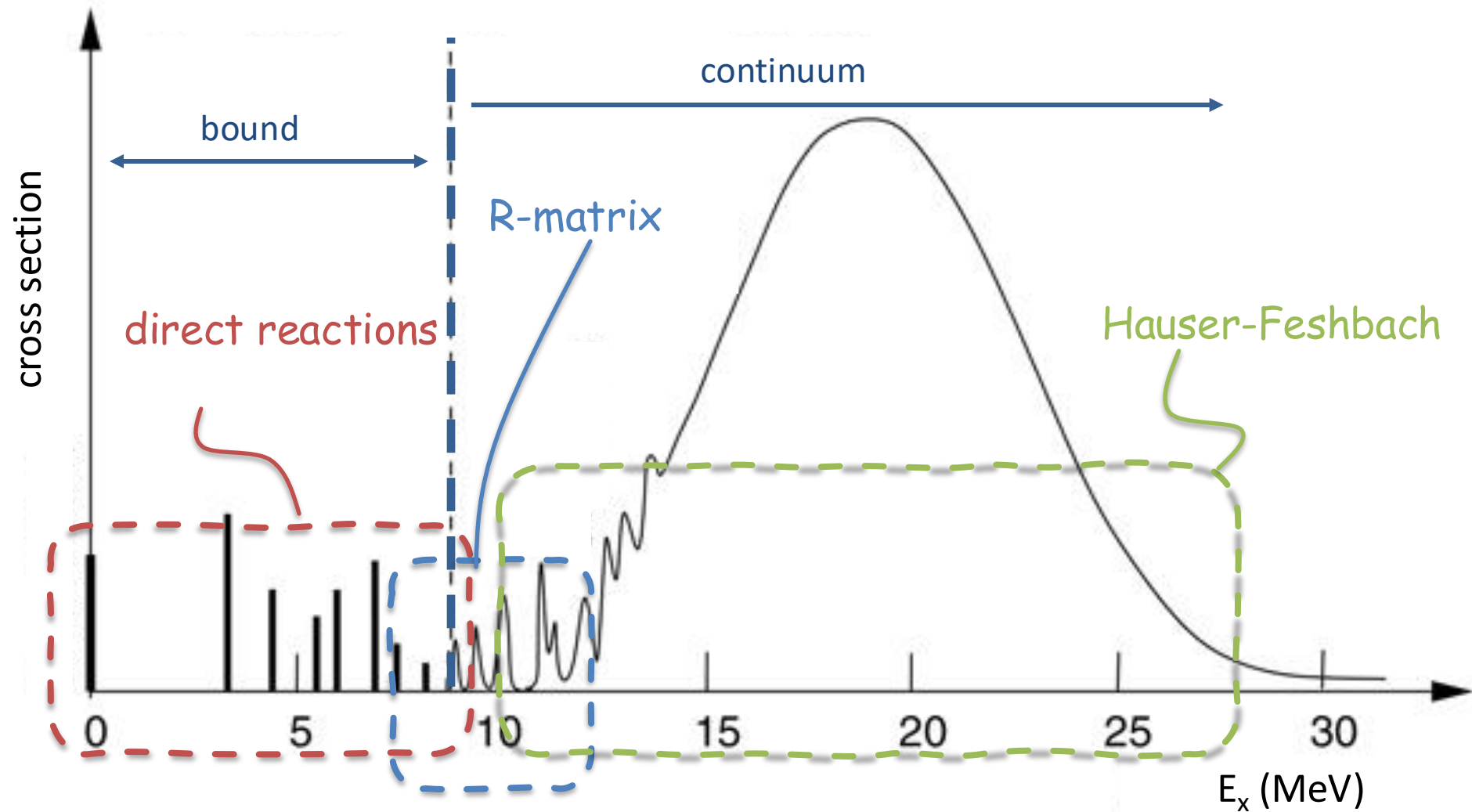
Which reaction theories, and where?



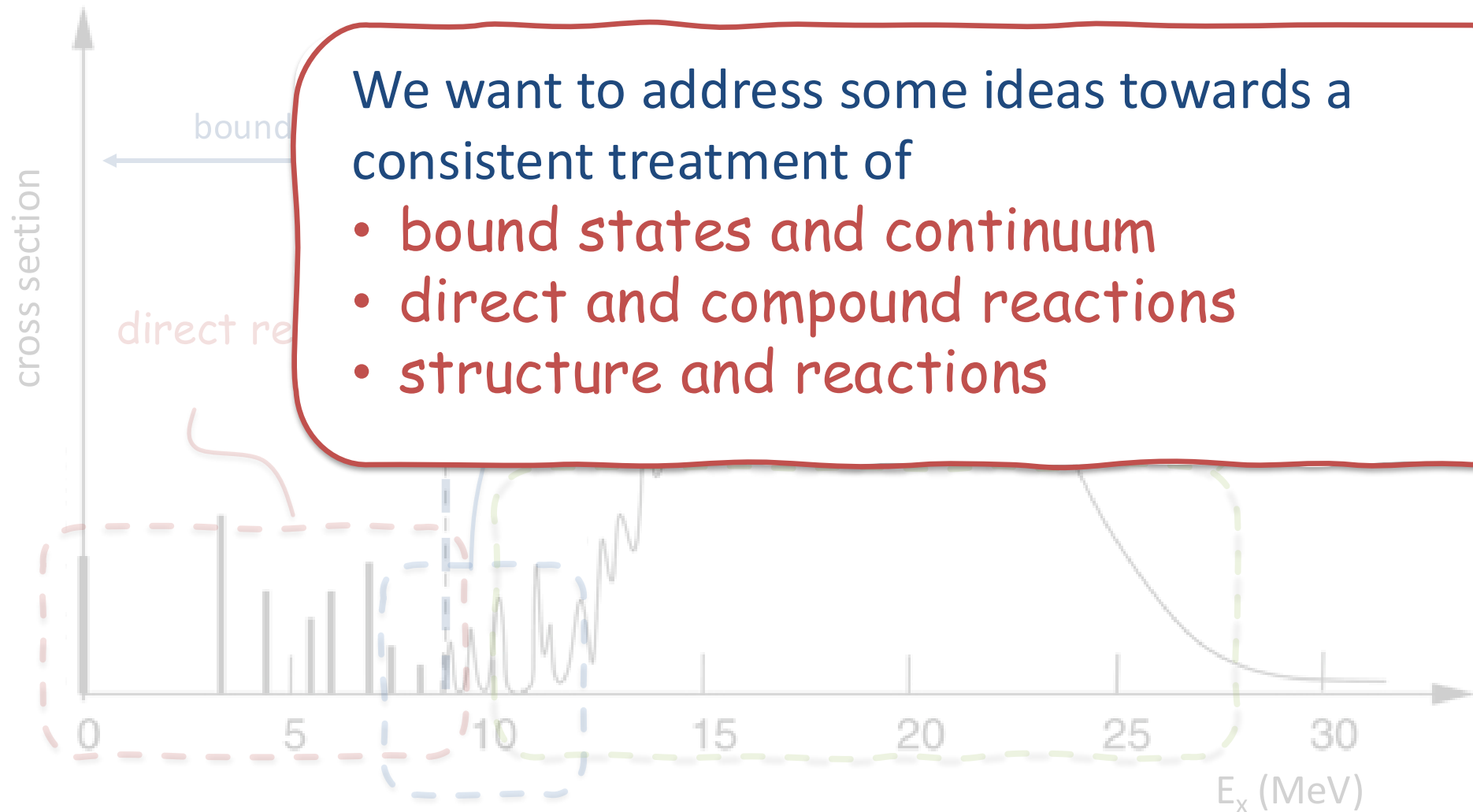
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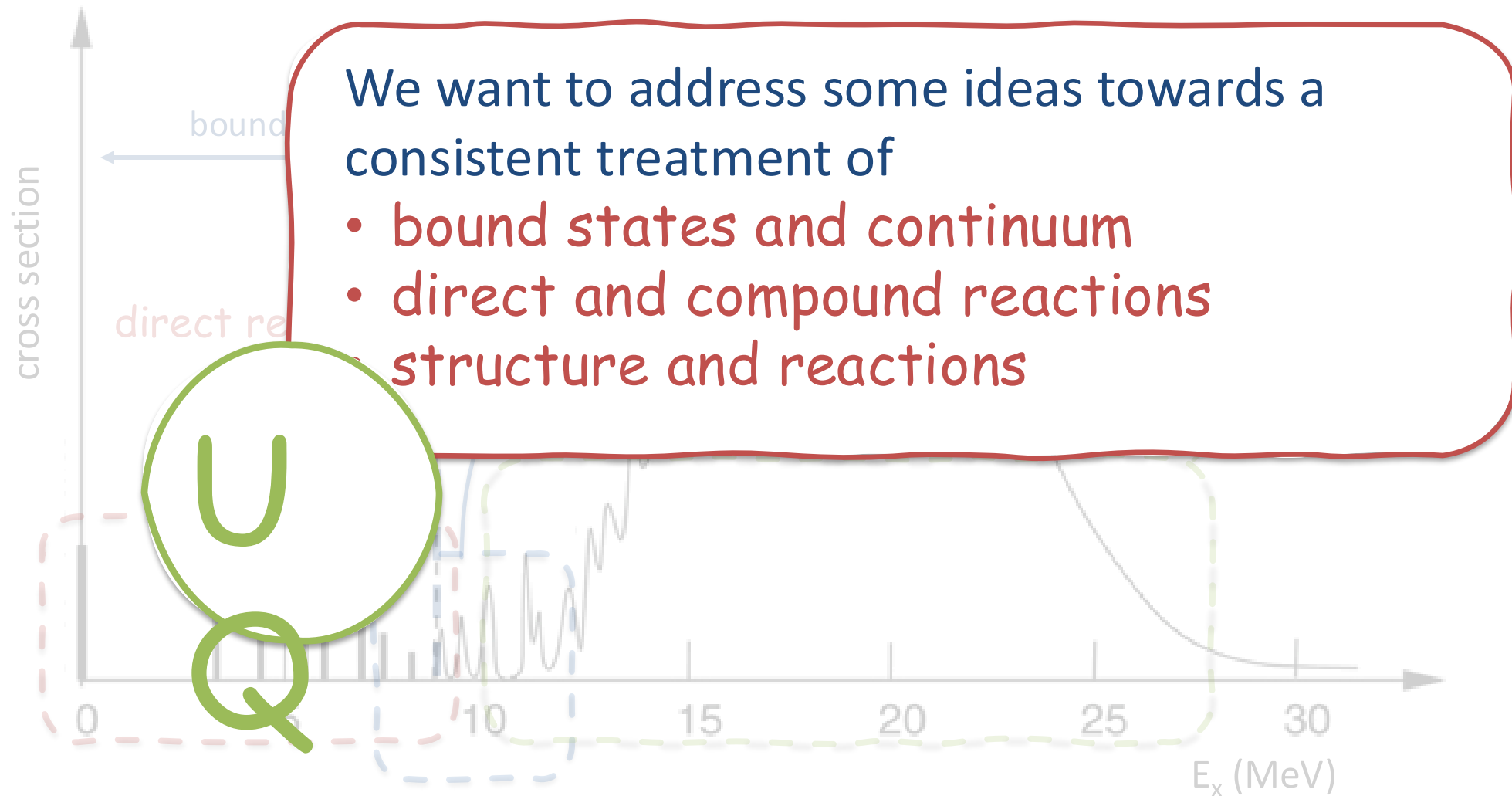
Which reaction theories, and where?



Can we treat everything on the same footing?

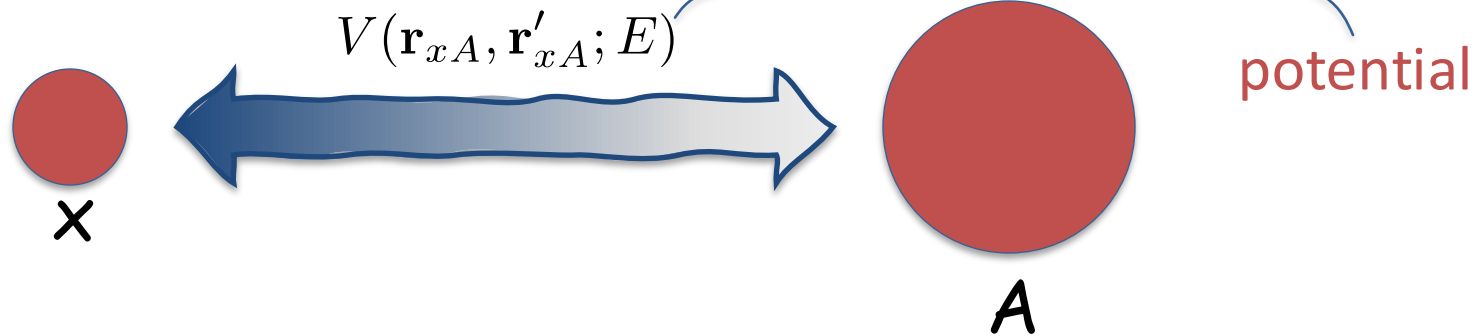


Can we treat everything on the same footing?



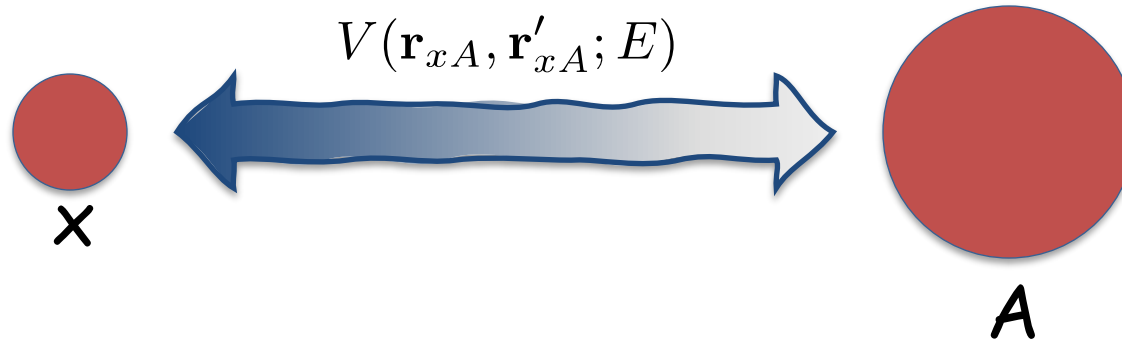
2-body scattering in a nutshell

elastic scattering between 2 nuclei x and A



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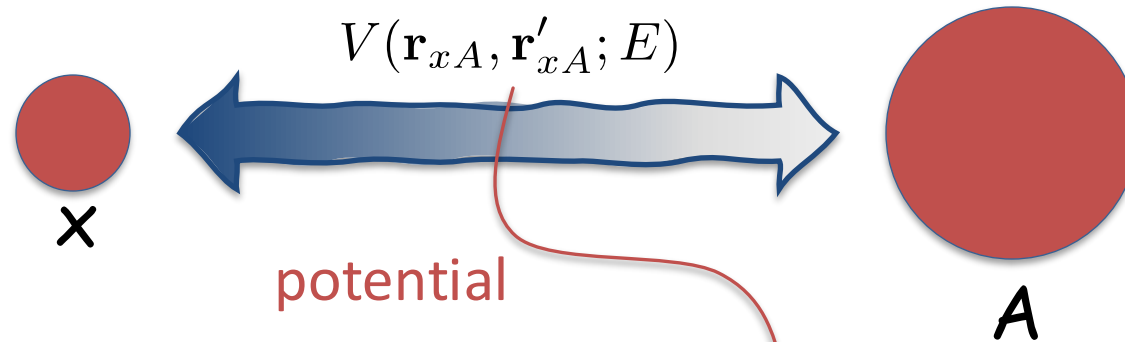


$$\psi_0(r_{xA}, E) = e^{i\mathbf{k}\mathbf{r}_{xA}} + G(E) V(E) e^{i\mathbf{k}\mathbf{r}_{xA}}$$

Lippmann-Schwinger equation

2-body scattering in a nutshell

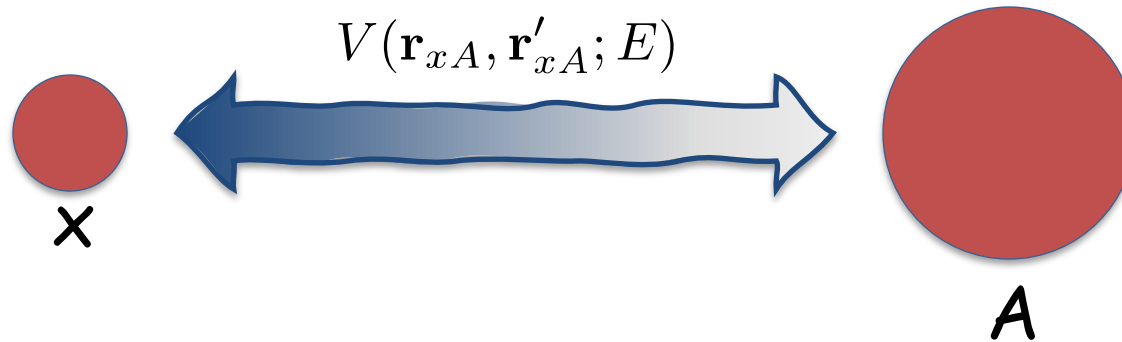
elastic scattering between 2 nuclei x and A



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2-body scattering in a nutshell

elastic scattering between 2 nuclei x and A



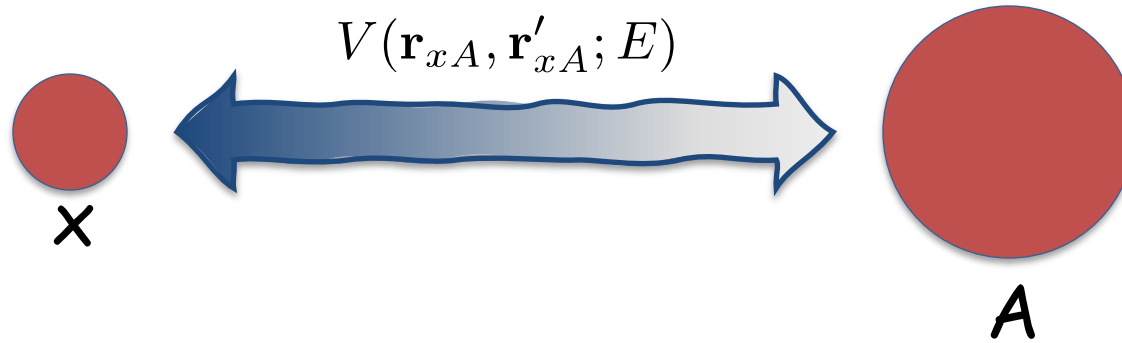
$$\psi_0(r_{xA}, E) = e^{i\mathbf{k}\mathbf{r}_{xA}} + G(E) V(E) e^{i\mathbf{k}\mathbf{r}_{xA}}$$

Green's function

$$G(E) = (E - T_x - V(E))^{-1}$$

2-body scattering in a nutshell

elastic scattering between 2 nuclei x and A



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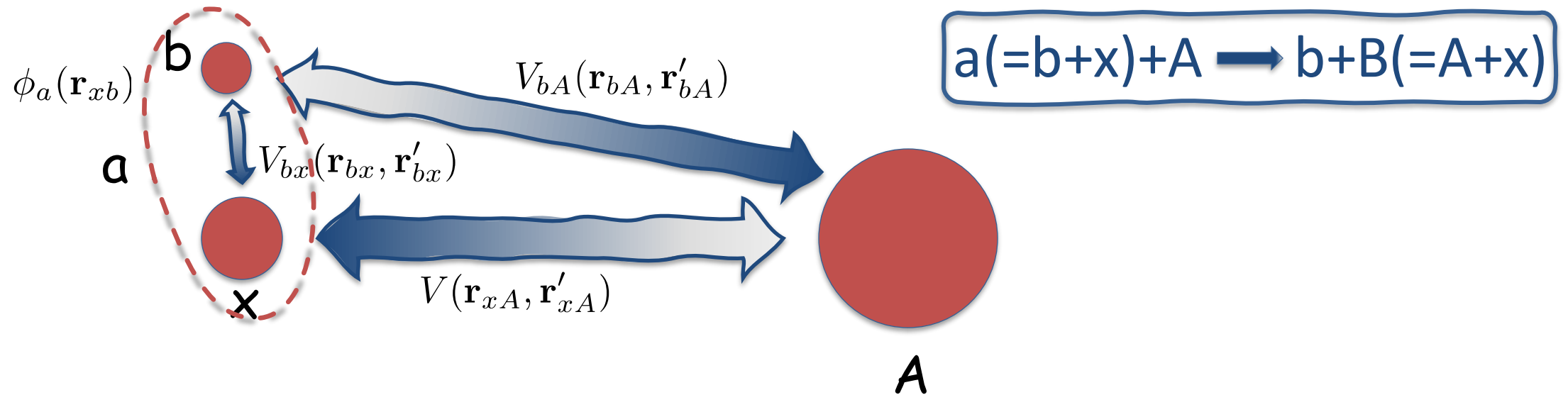
reaction cross section

$$\sigma_R = \frac{2\mu}{\hbar k_x} \langle \psi_0 | \text{Im} V | \psi_0 \rangle$$

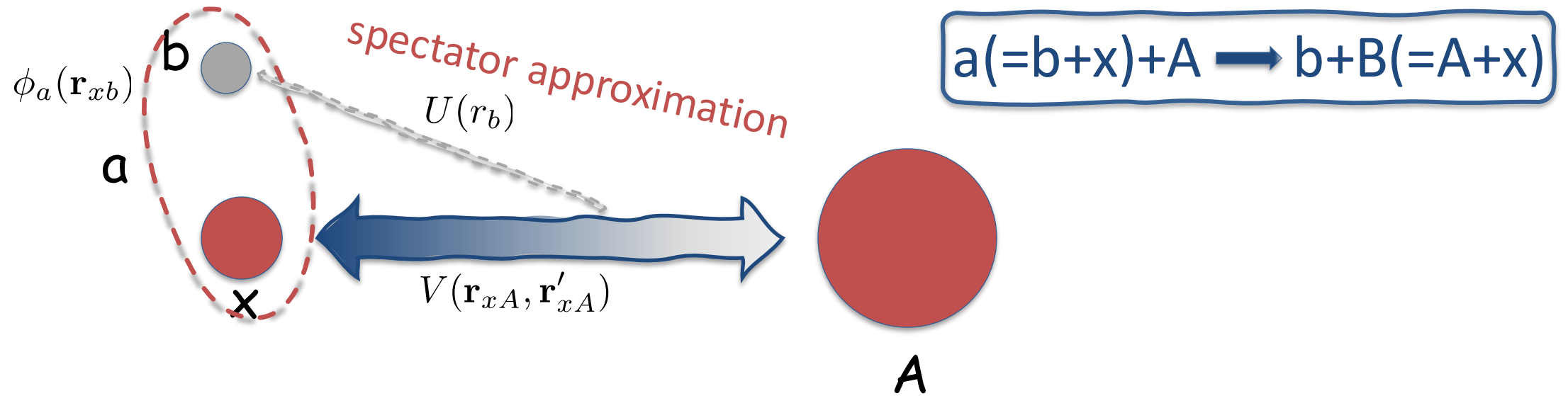
Part 2

Expanding the scope
to 3-body scattering: GFT

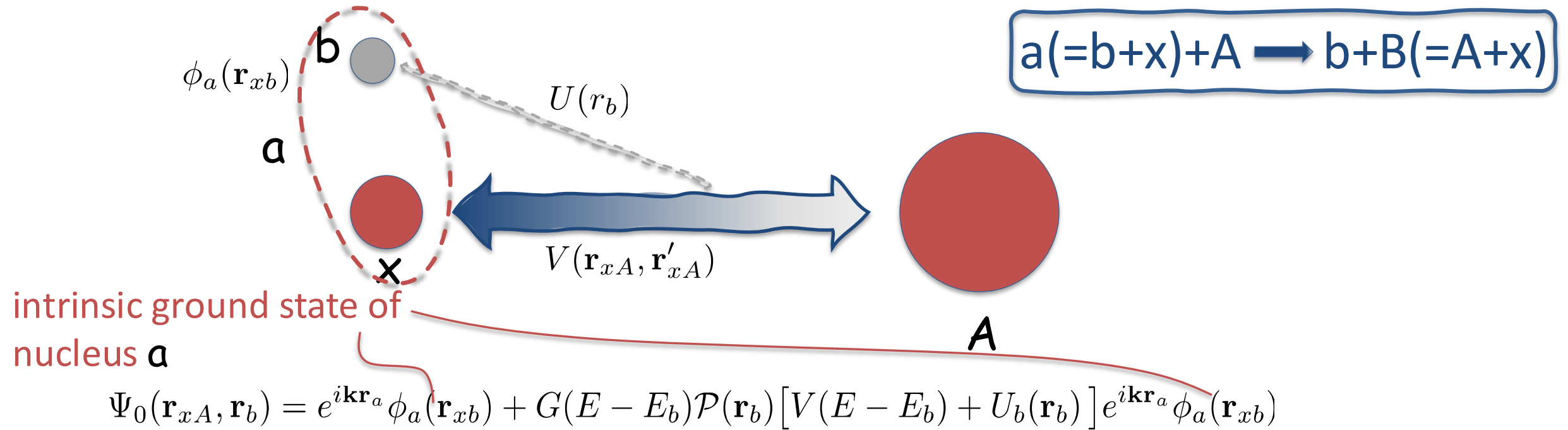
The Green's Function Transfer (GFT) formalism



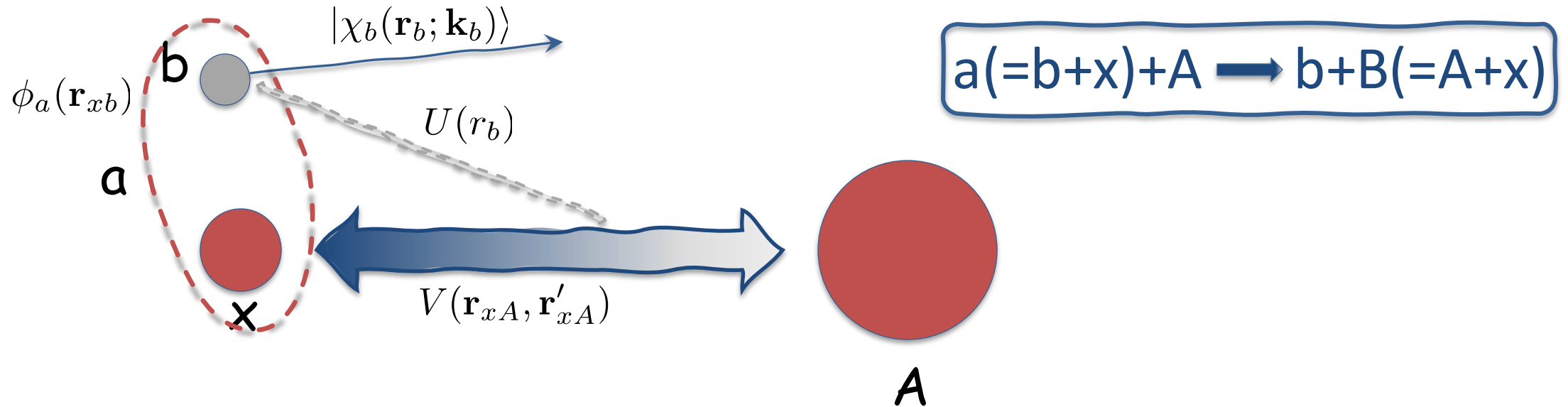
The Green's Function Transfer (GFT) formalism



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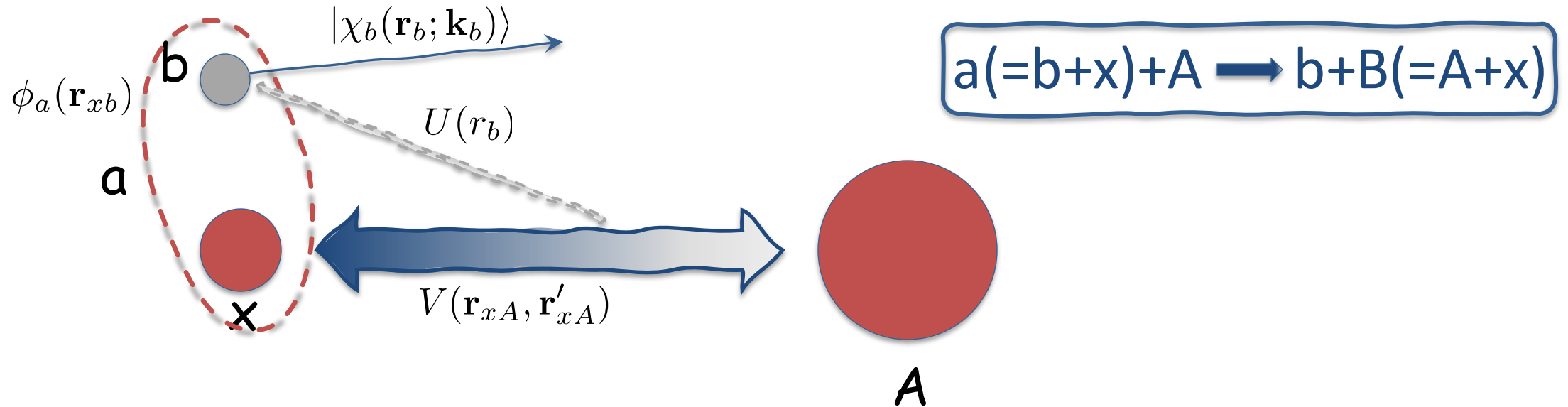


$$\Psi_0(\mathbf{r}_{xA}, \mathbf{r}_b) = e^{i\mathbf{k}\mathbf{r}_a} \phi_a(\mathbf{r}_{xb}) + G(E - E_b) \mathcal{P}(\mathbf{r}_b) [V(E - E_b) + U_b(r_b)] e^{i\mathbf{k}\mathbf{r}_a} \phi_a(\mathbf{r}_{xb})$$

projector over b states $\mathcal{P}(\mathbf{r}_b) = \int |\chi_b(\mathbf{r}_b; \mathbf{k}_b)\rangle \langle \chi_b(\mathbf{r}_b; \mathbf{k}_b)| d\mathbf{k}_b$

$$E_b = \frac{\hbar^2 k_b^2}{2\mu_b}$$

The Green's Function Transfer (GFT) formalism

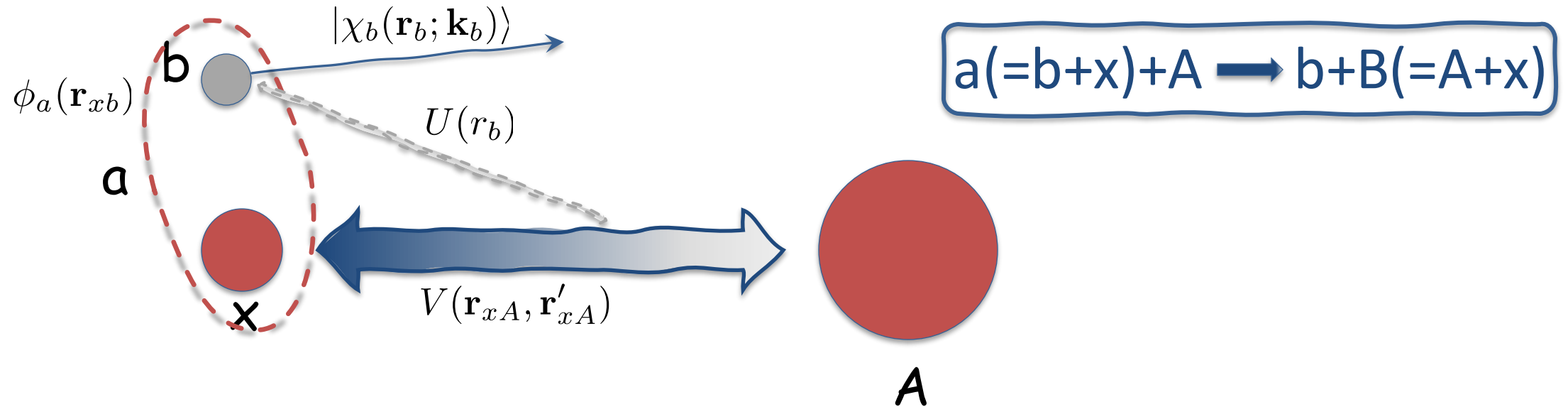


$$\Psi_0(\mathbf{r}_{xA}, \mathbf{r}_b) = e^{i\mathbf{k}\mathbf{r}_a} \phi_a(\mathbf{r}_{xb}) + G(E - E_b) \mathcal{P}(\mathbf{r}_b) [V(E - E_b) + U_b(\mathbf{r}_b)] e^{i\mathbf{k}\mathbf{r}_a} \phi_a(\mathbf{r}_{xb})$$

Green's function $G(E) = (E - T_x - V(E))^{-1}$

Same as for x - A scattering!

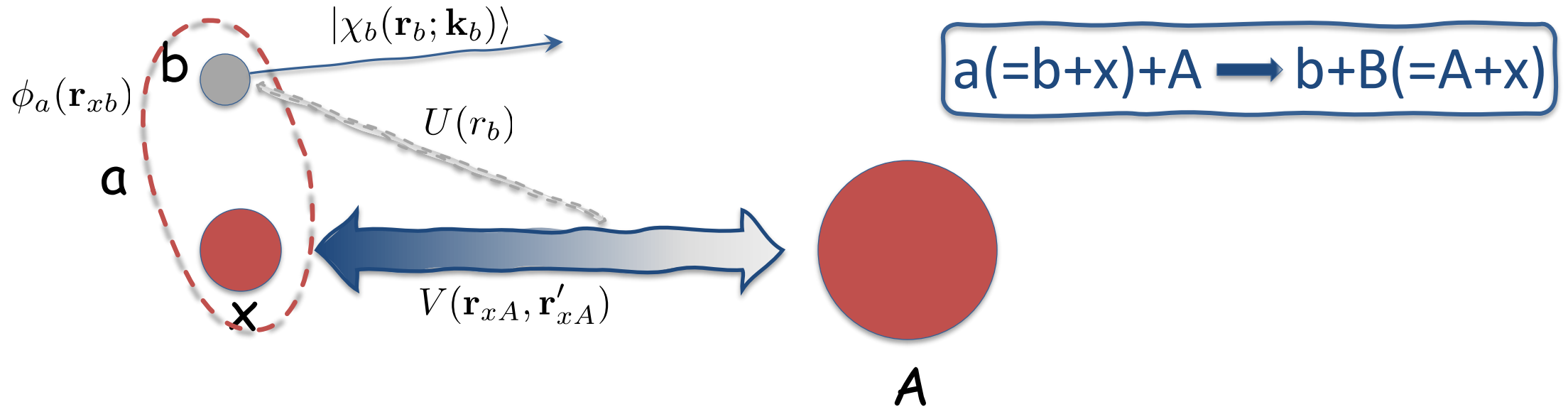
The Green's Function Transfer (GFT) formalism



$$\Psi_0(\mathbf{r}_{xA}, \mathbf{r}_b) = e^{i\mathbf{k}\mathbf{r}_a} \phi_a(\mathbf{r}_{xb}) + \boxed{G(E - E_b) \mathcal{P}(\mathbf{r}_b)} [V(E - E_b) + U_b(\mathbf{r}_b)] e^{i\mathbf{k}\mathbf{r}_a} \phi_a(\mathbf{r}_{xb})$$

factorized propagator

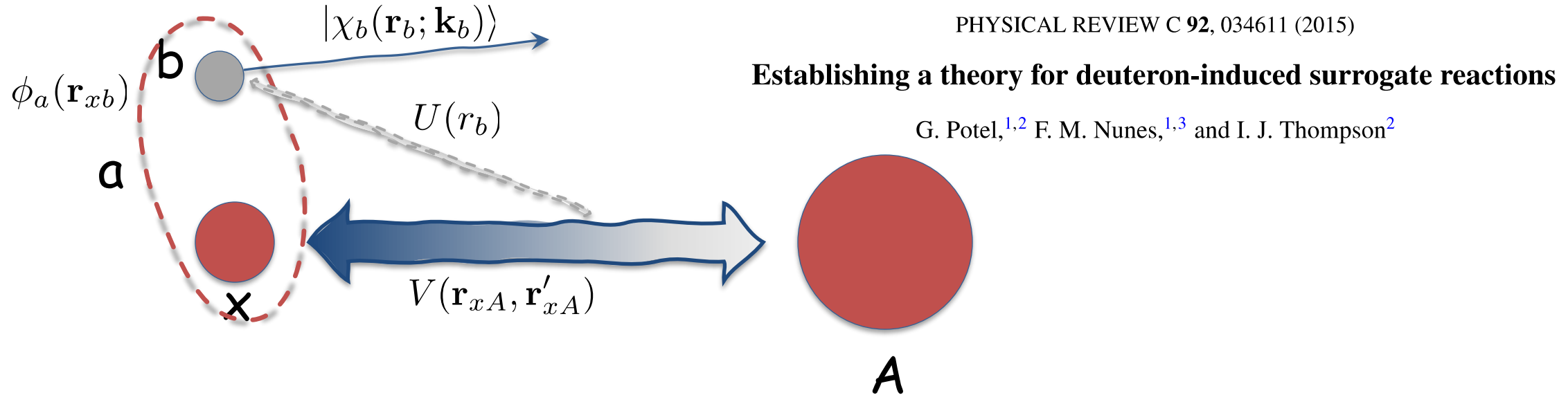
The Green's Function Transfer (GFT) formalism



$$\langle \chi_b(\mathbf{r}_b; \mathbf{k}_b) | \Psi_0(\mathbf{r}_{xA}, \mathbf{r}_b) = \langle \chi_b(\mathbf{r}_b; \mathbf{k}_b) | \left(e^{i\mathbf{k}\mathbf{r}_a} \phi_a(\mathbf{r}_{xb}) + G(E - E_b) \mathcal{P}(\mathbf{r}_b) [V(E - E_b) + U_b(\mathbf{r}_b)] e^{i\mathbf{k}\mathbf{r}_a} \phi_a(\mathbf{r}_{xb}) \right)$$

project over b state to get x - A wavefunction

The Green's Function Transfer (GFT) formalism



$$\psi_0^I(\mathbf{r}_{xA}, E) = \psi^{HM}(\mathbf{r}_a) + G(E - E_b) [V(E - E_b)\psi^{HM} + \langle \chi_b | U_b(\mathbf{r}_b)] e^{i\mathbf{k}\mathbf{r}_a} \phi_a(\mathbf{r}_{xb})$$

$$\psi^{HM}(\mathbf{r}_{xA}) = \int \chi_b^*(\mathbf{r}_{bB}, \mathbf{k}_b) \phi_a(\mathbf{r}_{xb}) e^{i\mathbf{k}\mathbf{r}_a} d\mathbf{r}_{xb} \quad \text{Hussein-McVoy term}$$

$$\sigma_R^I(E, E_b) = \frac{2\mu}{\hbar k_x} \langle \psi_0^I | \text{Im} \mathcal{V}(E - E_b) | \psi_0^I \rangle \quad \text{inclusive } \mathbf{x}\text{-}\mathbf{A} \text{ cross section}$$

Let us now introduce explicitly some **internal structure** ξ . Can we treat structure and reactions on the same footing?

$$\psi_0(r_{xA}, \xi, E) = e^{i\mathbf{k}\mathbf{r}_{xA}} \phi_0(\xi) + \mathbf{G}(E) \mathbf{V}(E) e^{i\mathbf{k}\mathbf{r}_{xA}} \phi_0(\xi)$$

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project on observed scattered direction



$$\langle e^{i\mathbf{k}_f\mathbf{r}_{xA}} \phi_0(\xi) | \psi_0(r_{xA}, \xi, E) \rangle = \langle e^{i\mathbf{k}_f\mathbf{r}_{xA}} \phi_0(\xi) | \mathbf{G}(E) \mathbf{V}(E) | e^{i\mathbf{k}\mathbf{r}_{xA}} \phi_0(\xi) \rangle$$

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optical potential

$$\langle e^{i\mathbf{k}_f\mathbf{r}_{xA}} \phi_0(\xi) | \psi_0(r_{xA}, \xi, E) \rangle = \langle e^{i\mathbf{k}_f\mathbf{r}_{xA}} \phi_0(\xi) | G(E) \mathcal{V}(E) | e^{i\mathbf{k}\mathbf{r}_{xA}} \phi_0(\xi) \rangle$$

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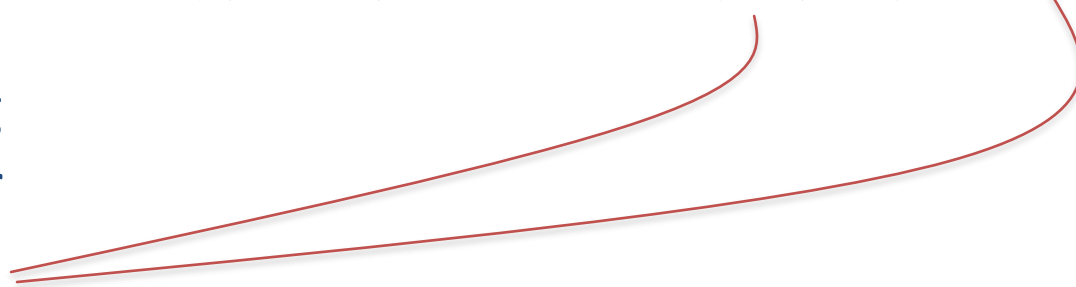
$$\psi_0(r_{xA}, \xi, E) = e^{i\mathbf{k}\mathbf{r}_{xA}} \phi_0(\xi) + \mathbf{G}(E) \mathbf{V}(E) e^{i\mathbf{k}\mathbf{r}_{xA}} \phi_0(\xi)$$



$$\langle e^{i\mathbf{k}_f \mathbf{r}_{xA}} \phi_0(\xi) | \psi_0(r_{xA}, \xi, E) \rangle = \langle e^{i\mathbf{k}_f \mathbf{r}_{xA}} \phi_0(\xi) | G(E) \mathcal{V}(E) | e^{i\mathbf{k}\mathbf{r}_{xA}} \phi_0(\xi) \rangle$$

classification according
to asymptotic behavior

$\lim_{r_{xA} \rightarrow \infty} = 0 \rightarrow \text{structure}$



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$\lim_{r_{xA} \rightarrow \infty} = 0 \rightarrow$ structure

$\lim_{r_{xA} \rightarrow \infty} = \text{oscillatory} \rightarrow$ reaction

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classification according
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$\lim_{r_{xA} \rightarrow \infty} = 0 \rightarrow \text{structure}$

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some possible implementations of
structure-reactions consistency

- ab-initio,
- EFT,
- QRPA+G-matrix,
- ...

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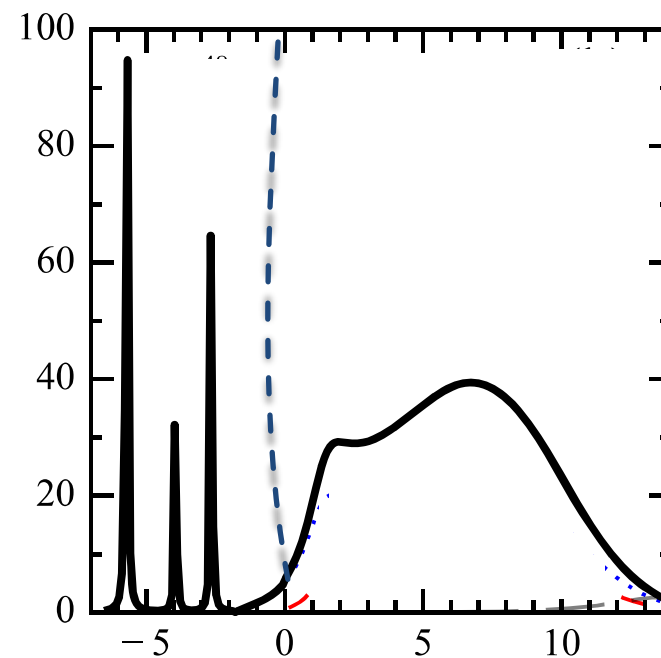
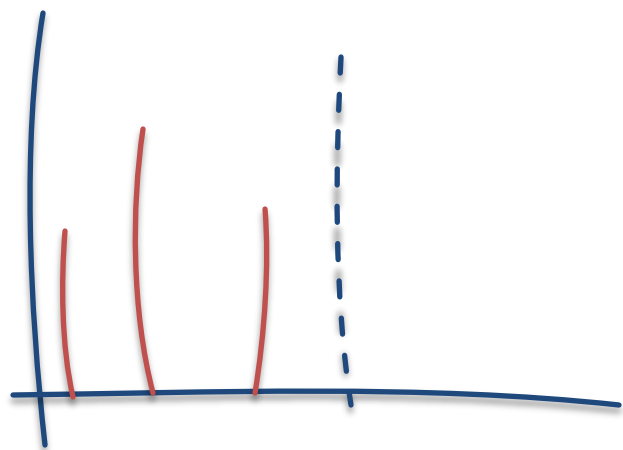
A proposed scheme to enforce structure-reactions consistency

$$\psi_i(r_{xA}, \xi) \rightarrow \mathcal{V}(E) \rightarrow G(E) = (E - T - \mathcal{V}(E))^{-1}$$

DWBA vs GFT

$$\sigma_{i0}^{DWBA} \sim |\langle \psi_i | V | \psi_0 \rangle|^2$$

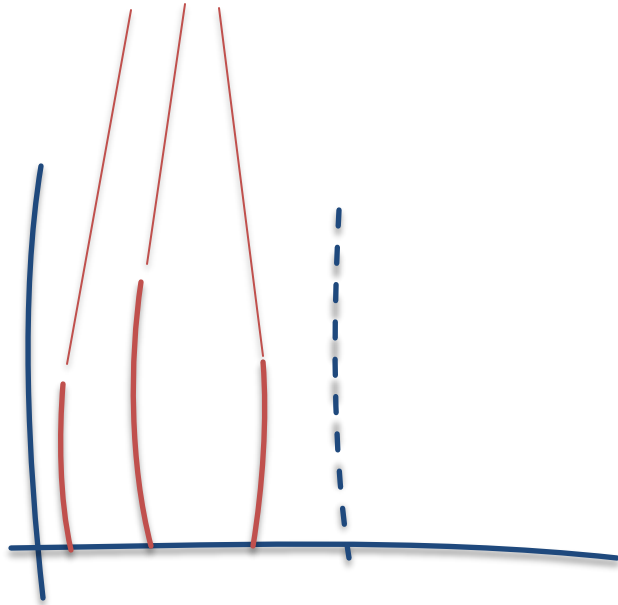
$$\sigma_R^{GFT}(E) \sim \langle G(E) (\mathcal{V}(E) + U_b) \psi^{HM} | \text{Im} \mathcal{V}(E) | G(E) (\mathcal{V}(E) + U_b) \psi^{HM} \rangle$$



DWBA vs GFT

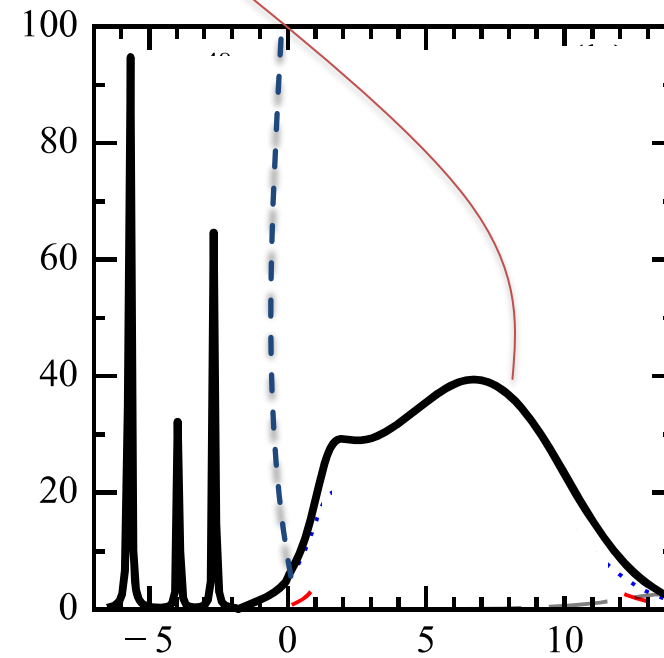
$$\sigma_{i0}^{DWBA} \sim |\langle \psi_i | V | \psi_0 \rangle|^2$$

discrete final states



$$\sigma_R^{GFT}(E) \sim \langle G(E) (\mathcal{V}(E) + U_b) \psi^{HM} | \text{Im} \mathcal{V}(E) | G(E) (\mathcal{V}(E) + U_b) \psi^{HM} \rangle$$

continuous function of E

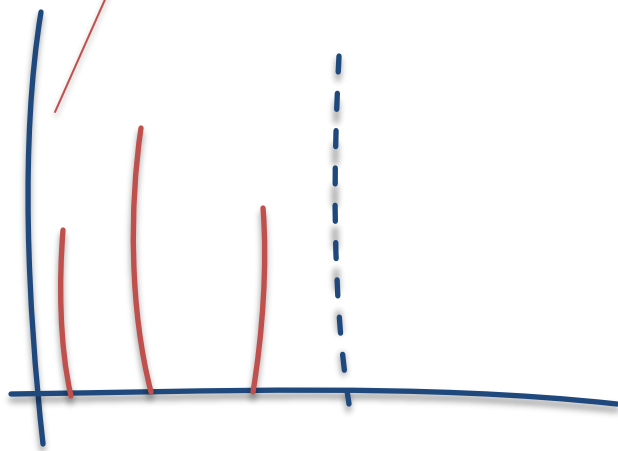


DWBA vs GFT

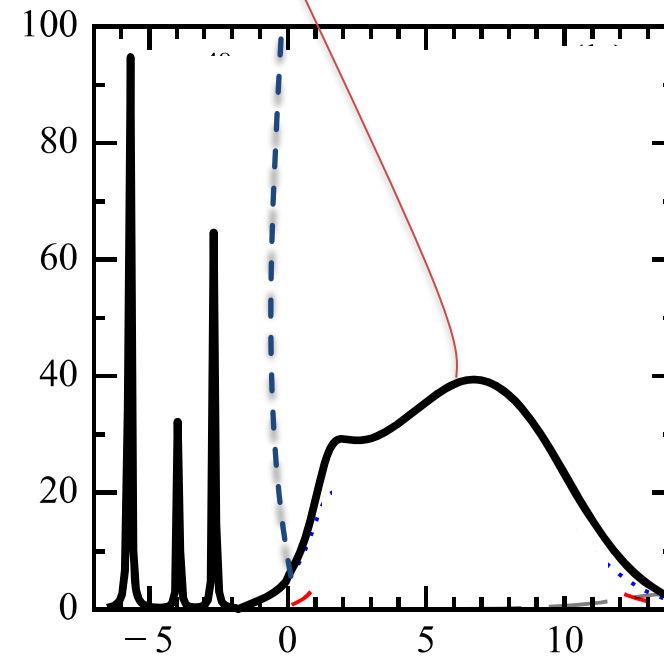
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extracted $S = \sigma_{\text{exp}} / \sigma_{\text{th}}$



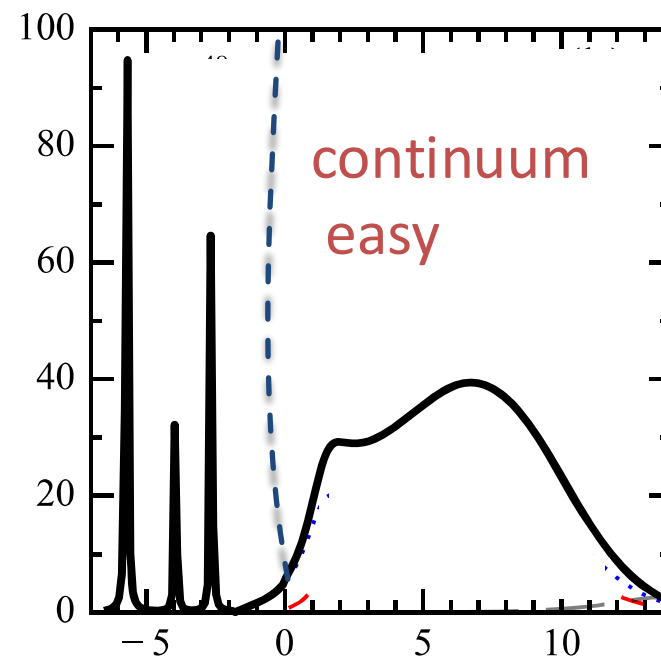
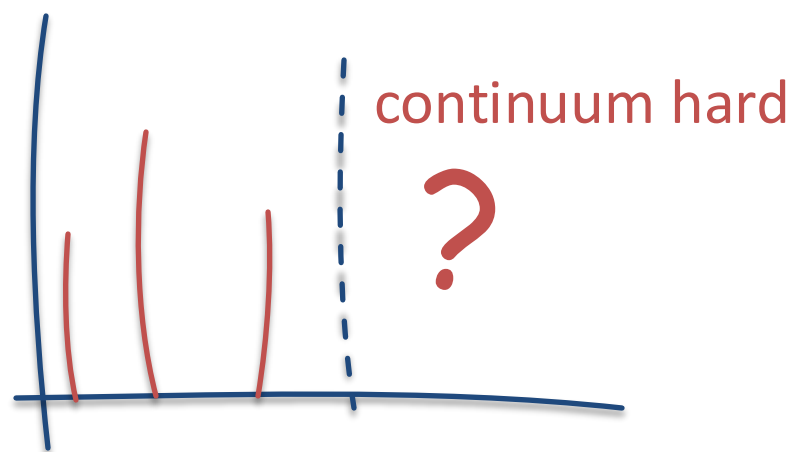
consistent normalization



DWBA vs GFT

$$\sigma_{i0}^{DWBA} \sim |\langle \psi_i | V | \psi_0 \rangle|^2$$

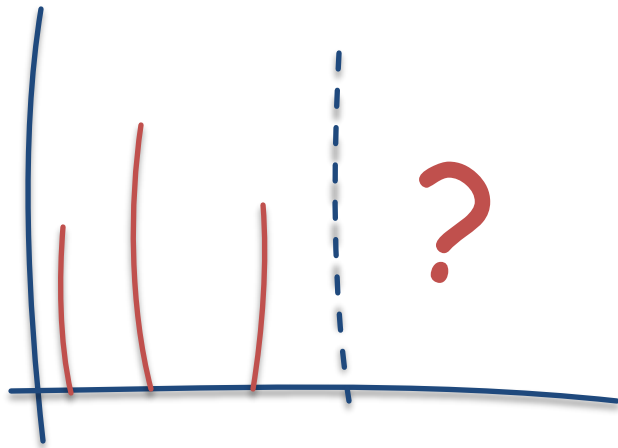
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DWBA vs GFT

$$\sigma_{i0}^{DWBA} \sim |\langle \psi_i | V | \psi_0 \rangle|^2$$

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$$G(E) = (E - T - \mathcal{V}(E))^{-1}$$

- Consistency between structure and reactions
- Same ingredients as χ -A scattering
- Need for tools for inverting Hamiltonians with non-local potentials

Part 3

Computing the Optical Potential

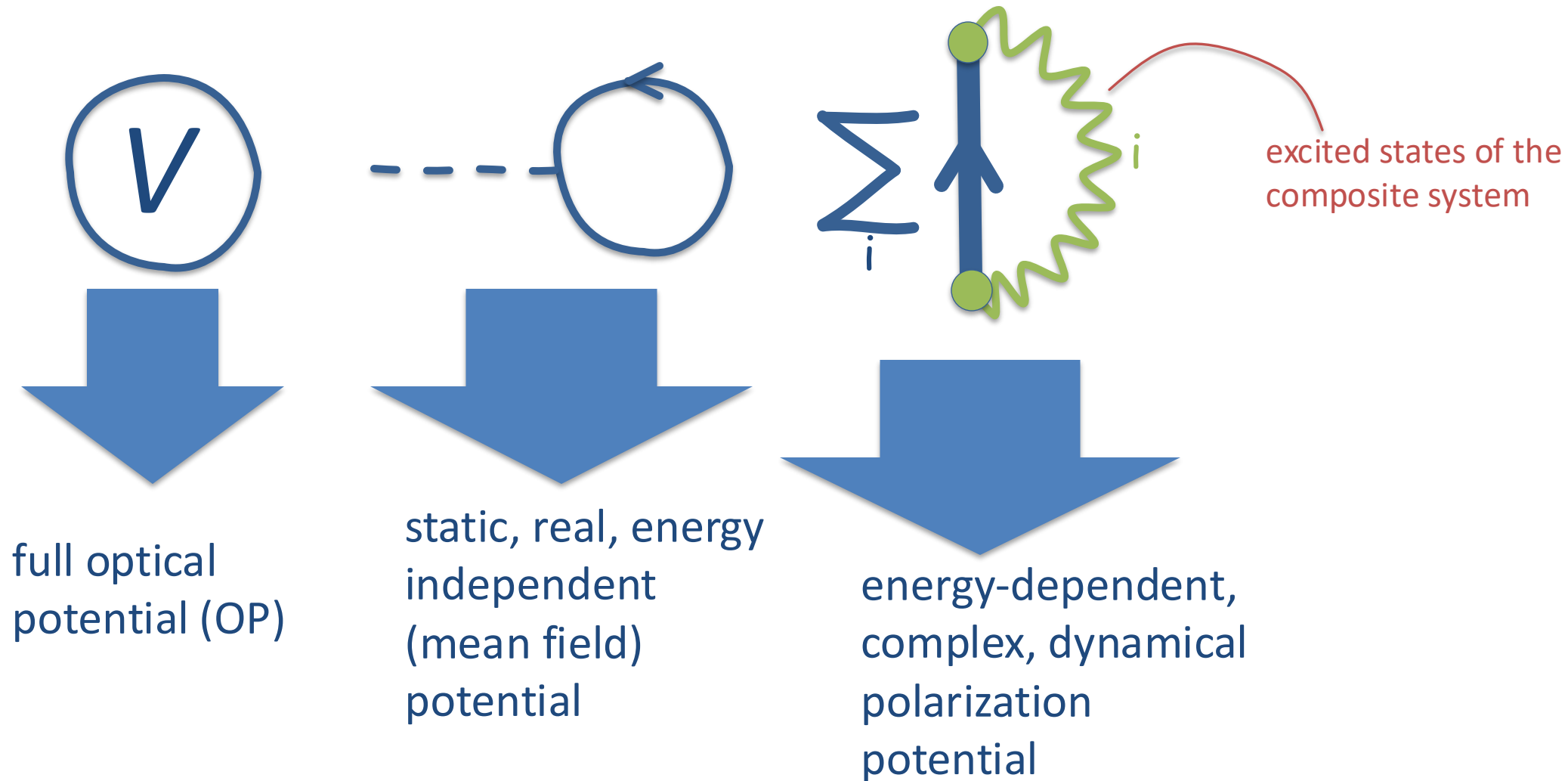
with G. Sargsyan

The Optical Potential is a projection of the many-body Hamiltonian on the elastic channel

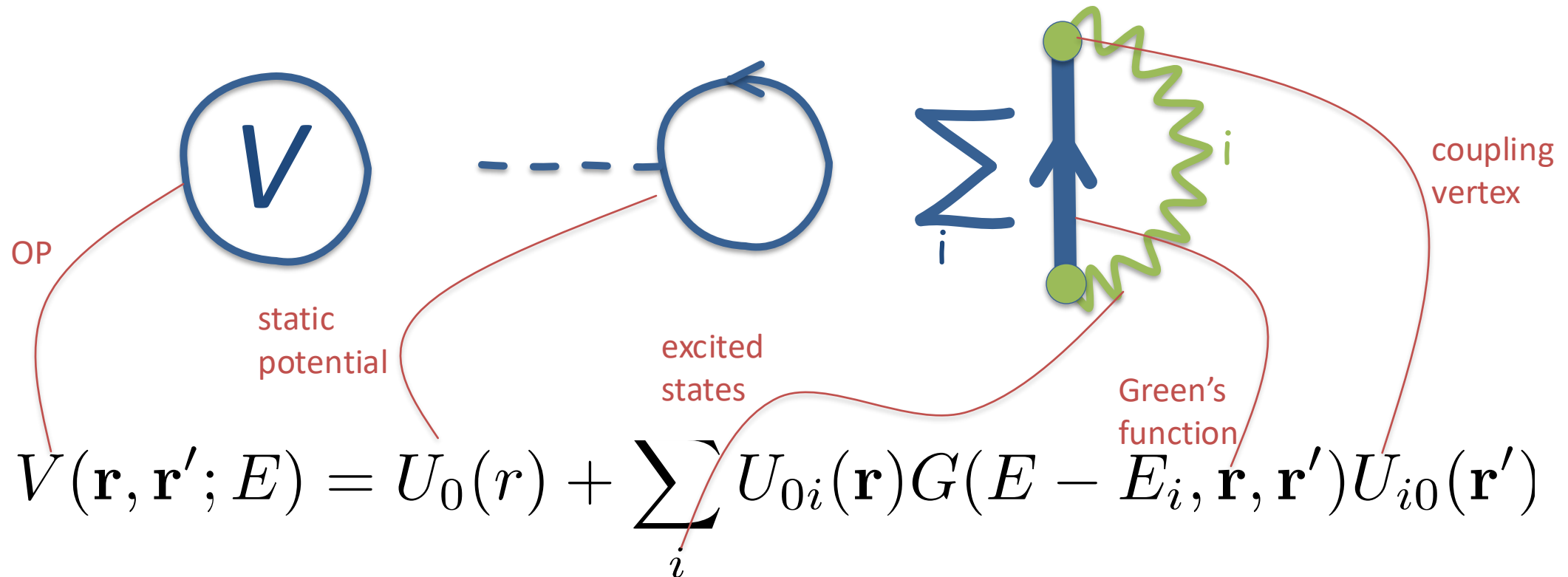
$$\begin{bmatrix} T+V_{00} & V_{01} & V_{02} & \bullet & \bullet & \bullet \\ V_{10} & T+V_{11} & V_{12} & \bullet & \bullet & \bullet \\ V_{20} & V_{21} & T+V_{22} & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & & \bullet & \\ \bullet & \bullet & \bullet & & & \bullet \end{bmatrix} \Rightarrow \left[\textcircled{V} \right]$$

- The “**optical reduction**” transforms a many-body operator into a one-body operator
- It is a **well-defined**, in principle **exact**, mathematical operation

The OP accounts for the composite nature of the target nucleus



The OP accounts for the composite nature of the target nucleus



- The computed OP is energy dependent, non-local, complex, and dispersive
- The OP verifies the Kramers-Kronig dispersion relations between the real and the imaginary part

$^{24}\text{Mg}+n$ with valence shell model

step 1: ^{25}Mg shell-model calculation

excitation energy E_i

angular momentum

parity

spectroscopic
factor S_i

0	0	2	1	0.584066
0.701	0	0	1	-0.716831
1.169	0	2	1	0.488705
2.033	0	2	1	-0.288311
2.529	0	0	1	0.318332
2.701	0	2	1	0.542416
3.859	0	2	1	0.0495903
3.926	0	1	-1	-0.0298132
4.118	0	1	-1	-0.584623
4.226	0	3	-1	-0.651056
4.46	0	2	1	0.100777
4.816	0	2	1	0.0975601
4.945	0	1	-1	0.516883
5.065	0	1	-1	-0.0511968
5.416	0	0	1	-0.283037
5.638	0	3	-1	-0.219064
5.785	0	2	1	-0.0103979
5.792	0	2	1	0.132477
5.935	0	3	-1	0.069005
5.936	0	2	1	0.094658
6.033	0	1	-1	0.0353684
6.12	0	1	-1	-0.159821
6.243	0	2	1	-0.174362
6.35	0	3	-1	0.122727
6.385	0	0	1	0.182001
6.417	0	2	1	0.115995
6.609	0	2	1	0.100457
6.739	0	3	-1	0.157325
6.771	0	1	-1	0.419452
6.801	0	3	-1	0.160889

~600 states from $E_i=0$ to $E_i=14.6$ MeV

Shell model calculations by [K. Kravvaris](#)
with PSDPF interaction M Bouhelal, *et al.*, Nucl. Phys. A **864** (2011)

$^{24}\text{Mg}+n$ with valence shell model

step 2: Static potential and couplings

excitation energy E_i

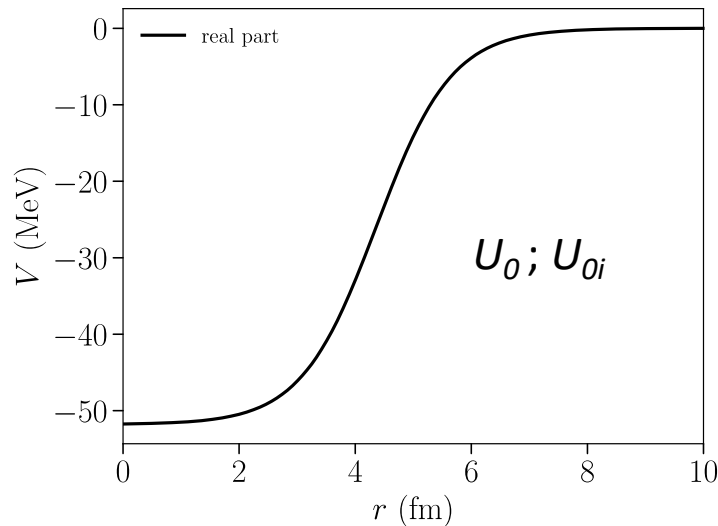
angular momentum

parity

spectroscopic factor S_i

0	0	2	1	0.584066
0.701	0	0	1	-0.716831
1.169	0	2	1	0.488705
2.033	0	2	1	-0.288311
2.529	0	0	1	0.318332
2.701	0	2	1	0.542416
3.859	0	2	1	0.0495903
3.926	0	1	-1	-0.0298132
4.118	0	1	-1	-0.584623
4.226	0	3	-1	-0.651056
4.46	0	2	1	0.100777
4.816	0	2	1	0.0975601
4.945	0	1	-1	0.516883
5.065	0	1	-1	-0.0511968
5.416	0	0	1	-0.283037
5.638	0	3	-1	-0.219064
5.785	0	2	1	-0.0103979
5.792	0	2	1	0.132477
5.935	0	3	-1	0.069005
5.936	0	2	1	0.094658
6.033	0	1	-1	0.0353684
6.12	0	1	-1	-0.159821
6.243	0	2	1	-0.174362
6.35	0	3	-1	0.122727
6.385	0	0	1	0.182001
6.417	0	2	1	0.115995
6.609	0	2	1	0.100457
6.739	0	3	-1	0.157325
6.771	0	1	-1	0.419452
6.801	0	3	-1	0.160889

$$V(\mathbf{r}, \mathbf{r}'; E) = U_0(r) + \sum_i U_{0i}(\mathbf{r}) G(E - E_i, \mathbf{r}, \mathbf{r}') U_{i0}(\mathbf{r}')$$



- static potential U_0 : real, local Woods-Saxon adjusted to reproduce binding energy of ^{25}Mg
- couplings U_{0i} : same Woods-Saxon, but adjusted to each E_i and multiplied by spectroscopic factor S_i

$^{24}\text{Mg}+n$ with valence shell model

step 2: Static potential and couplings

excitation energy E_i

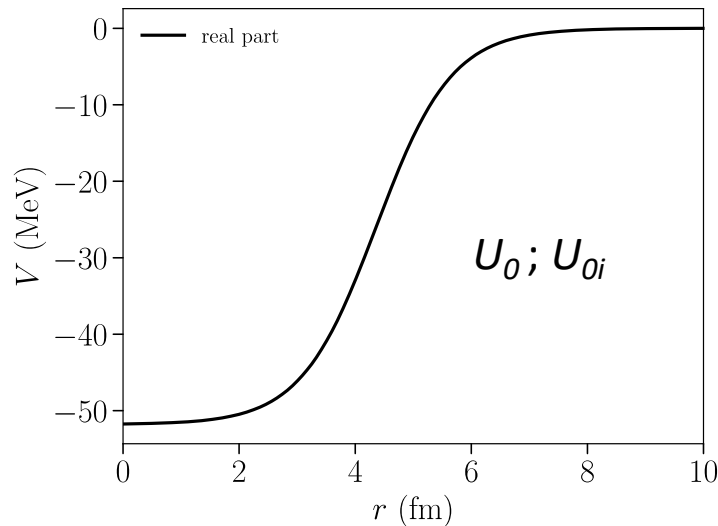
angular momentum

parity

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$$V(\mathbf{r}, \mathbf{r}'; E) = U_0(r) + \sum_i U_{0i}(\mathbf{r}) G(E - E_i, \mathbf{r}, \mathbf{r}') U_{i0}(\mathbf{r}')$$



- static potential U_0 : real, local Woods-Saxon adjusted to reproduce binding energy of ^{25}Mg
- couplings U_{0i} : same Woods-Saxon, but adjusted to each E_i and multiplied by spectroscopic factor S_i

can be done better!



$^{24}\text{Mg}+n$ with valence shell model

step 3: Iterative procedure

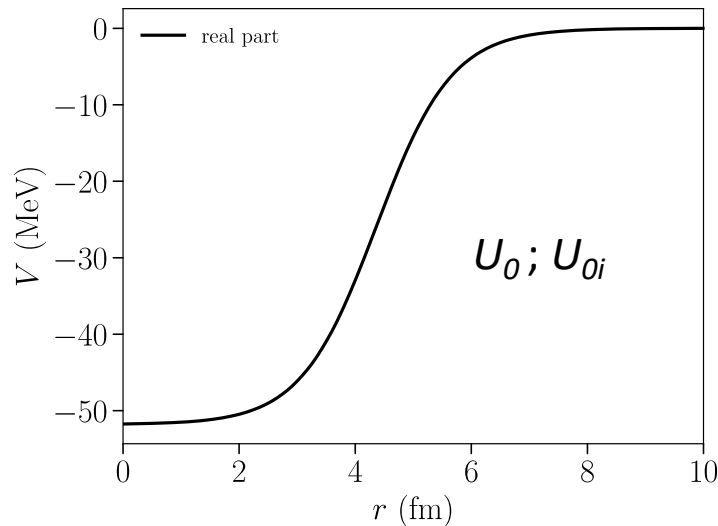
excitation energy E_i

angular momentum

parity

spectroscopic factor S_i

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1.169	0	2	1	0.488705
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6.801	0	3	-1	0.160889



$$V(\mathbf{r}, \mathbf{r}'; E) = U_0(r) + \sum_i U_{0i}(\mathbf{r}) G(E - E_i, \mathbf{r}, \mathbf{r}') U_{i0}(\mathbf{r}')$$

$$G(\mathbf{r}, \mathbf{r}', E) = (E - T - V(\mathbf{r}, \mathbf{r}'; E))^{-1}$$

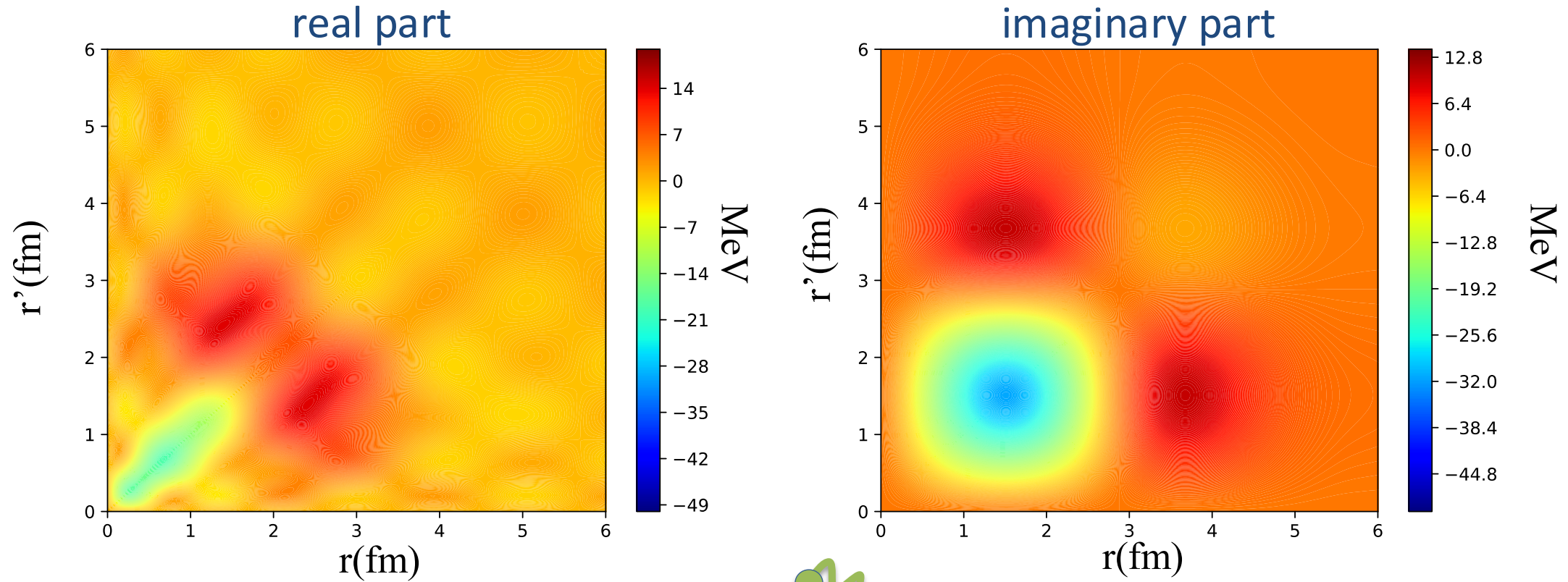
- Iterate until **convergence** is achieved
- Consistency between **potential** and **Green's** function is achieved, as expressed by **Dyson's** equation:

$$G(\mathbf{r}, \mathbf{r}'; E) = G_0(\mathbf{r}, \mathbf{r}'; E) + G_0(\mathbf{r}, \mathbf{r}'; E) V(\mathbf{r}, \mathbf{r}'; E) G(\mathbf{r}, \mathbf{r}'; E)$$

$$G_0(\mathbf{r}, \mathbf{r}'; E) = (E - T - U_0(r))^{-1}$$

As a bonus, we obtain the Green's function

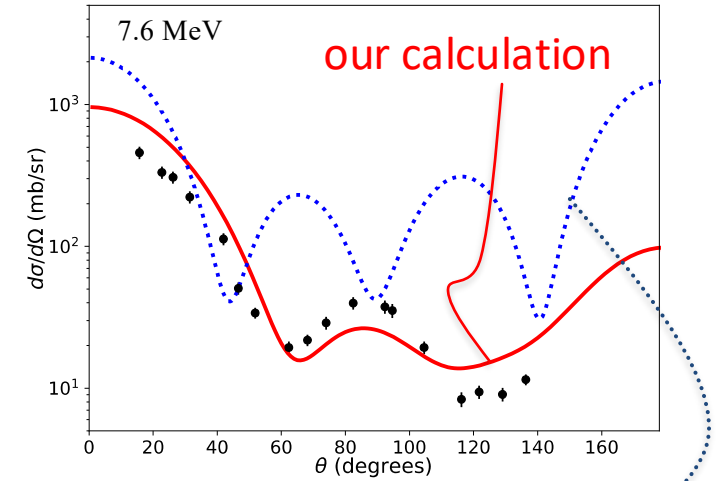
The dynamical polarization potential is complex, energy-dependent, dispersive, and non-local



$$V_{pol}(r, r'; E) = \sum_i \text{[diagram of a nucleon with a wavy line]} i ; \text{ for } E=1\text{MeV}$$

(see G. Blanchon talk)

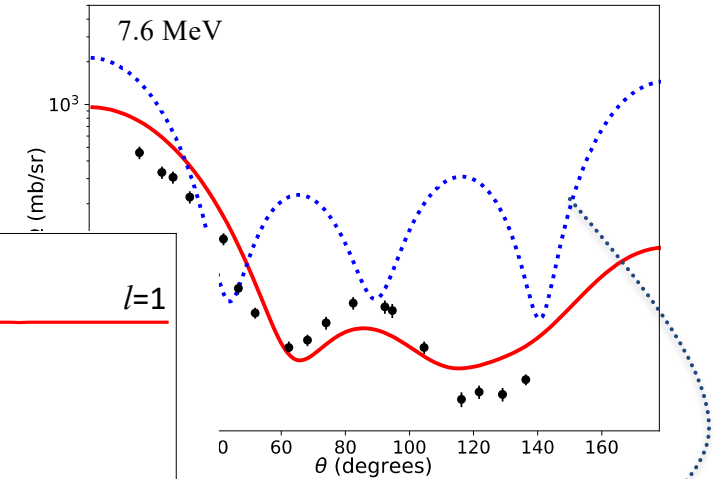
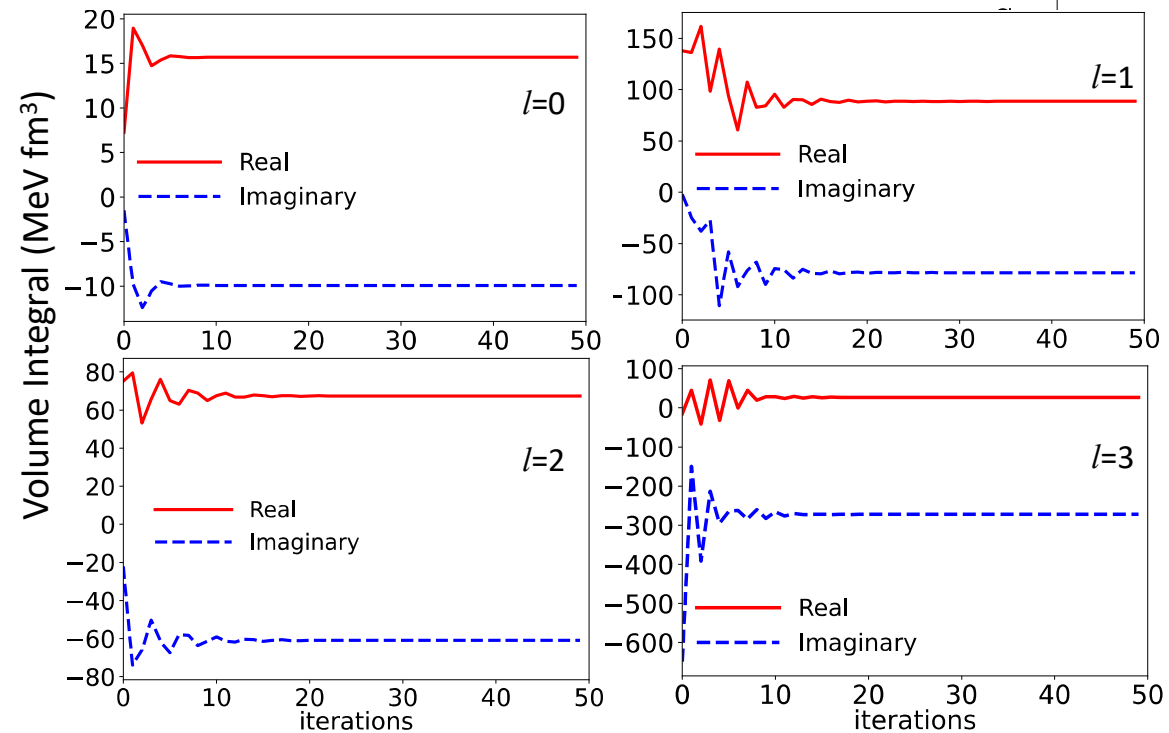
Our ^{24}Mg calculation compares well with experiment



not surprisingly, the static potential alone gives a very wrong result!

Our ^{24}Mg calculation compares well with experiment

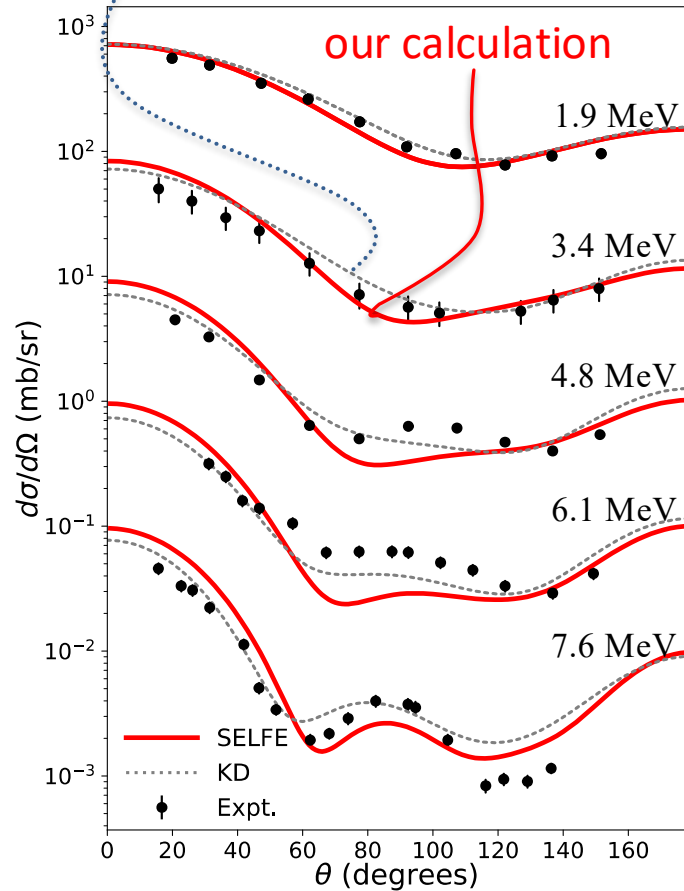
we check for convergence by looking at the volume integrals as a function of the iteration



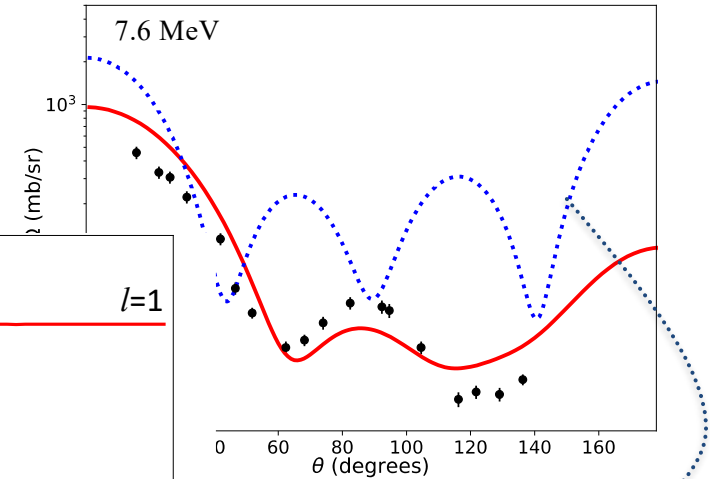
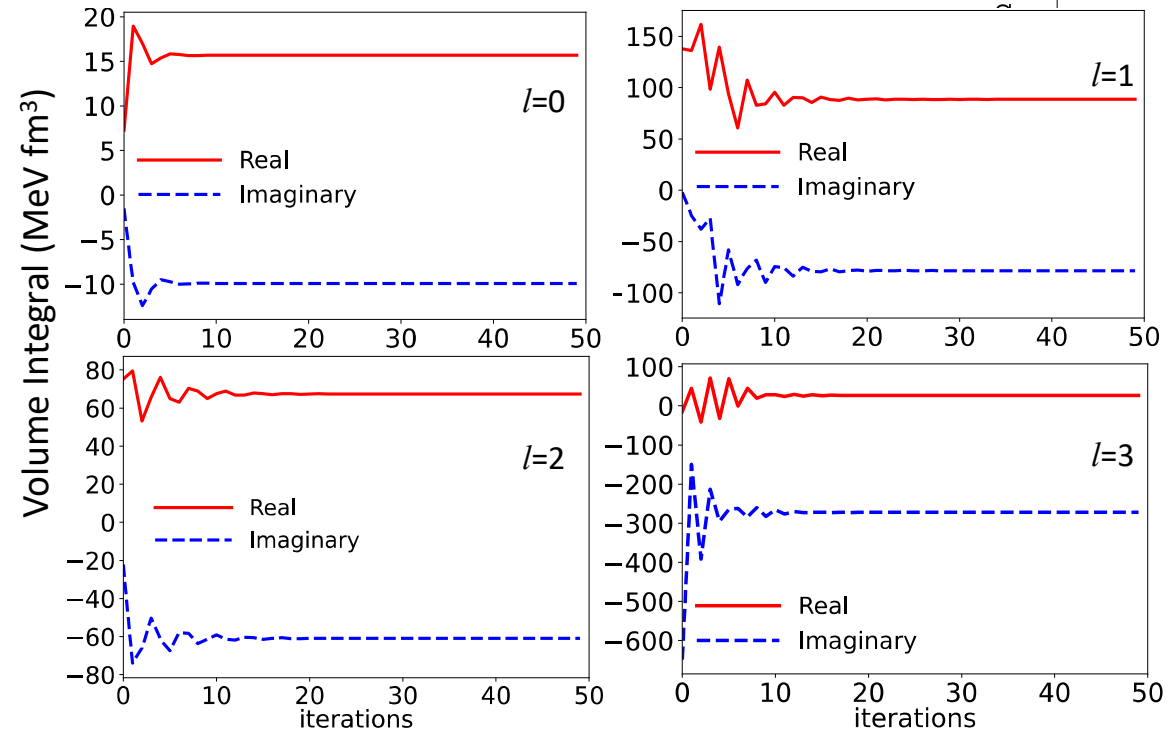
not surprisingly, the static potential alone gives a very wrong result!

Our ^{24}Mg calculation compares well with experiment

Koning-Delaroche



we check for convergence by looking at the volume integrals as a function of the iteration



not surprisingly, the static potential alone gives a very wrong result!

Part 4

Applying the GFT

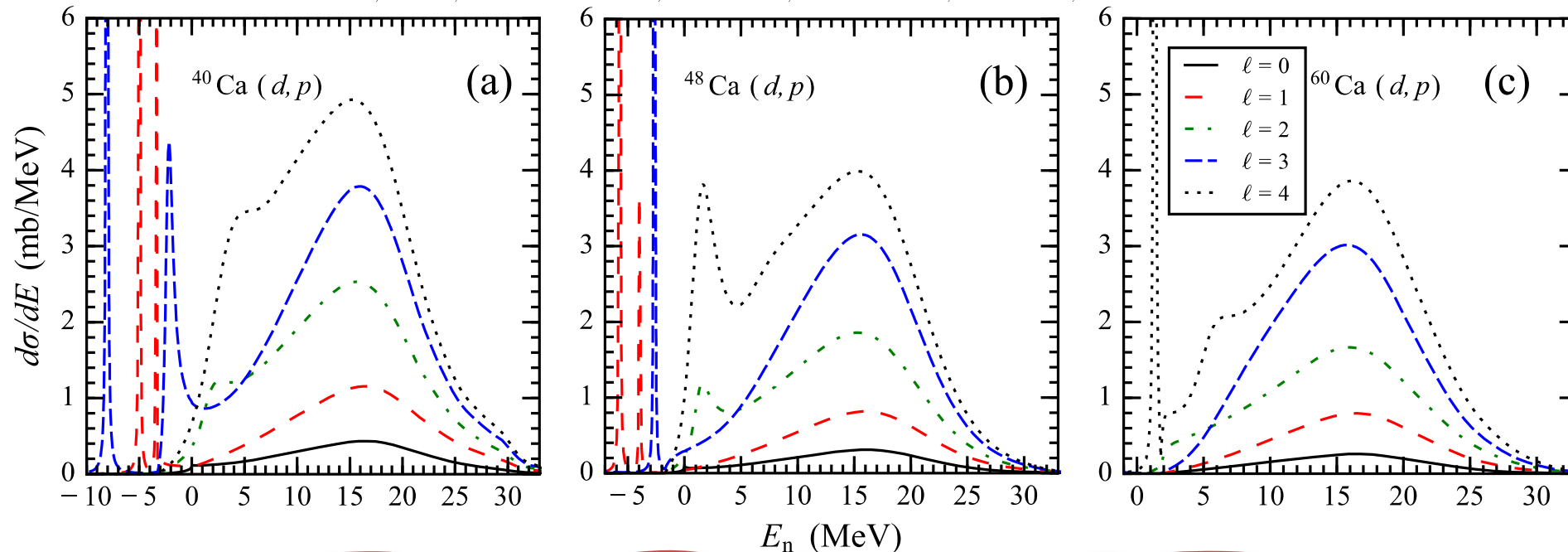
GFT with the Dispersive Optical Model (DOM); Ca(d,p)

Eur. Phys. J. A (2017) 53: 178

THE EUROPEAN
PHYSICAL JOURNAL A

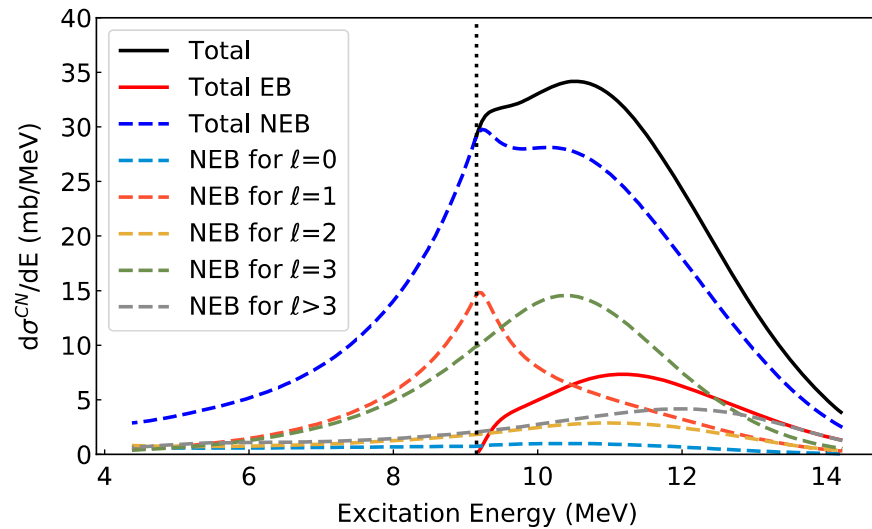
Toward a complete theory for predicting inclusive deuteron breakup away from stability

G. Potel^{1,a}, G. Perdikakis^{1,2,3,b}, B.V. Carlson^{4,c}, M.C. Atkinson⁵, W.H. Dickhoff⁵, J.E. Escher⁶, M.S. Hussein^{4,7,8}, J. Lei^{9,d}, W. Li¹, A.O. Macchiavelli¹⁰, A.M. Moro⁹, F.M. Nunes^{1,11}, S.D. Pain¹², and J. Rotureau¹



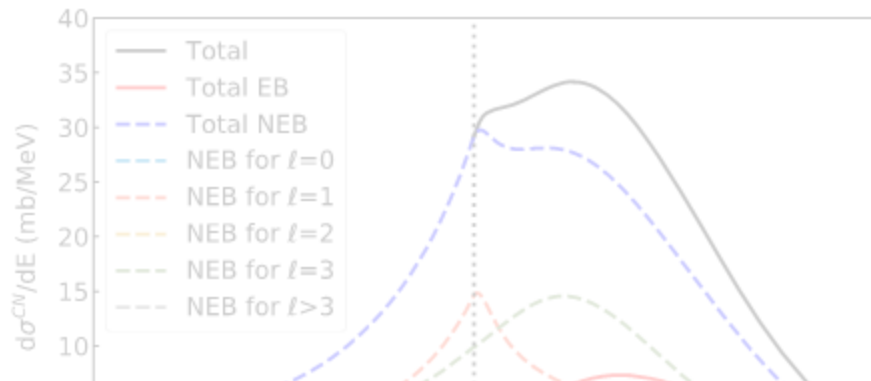
- Dispersive: reproduction of positive and negative energy cross section
- Controlled extrapolation to exotic nuclei

GFT applied to the Surrogate Method: $^{95}\text{Mo}(\text{d},\text{p}\gamma)$ with Koning-Delaroche OP

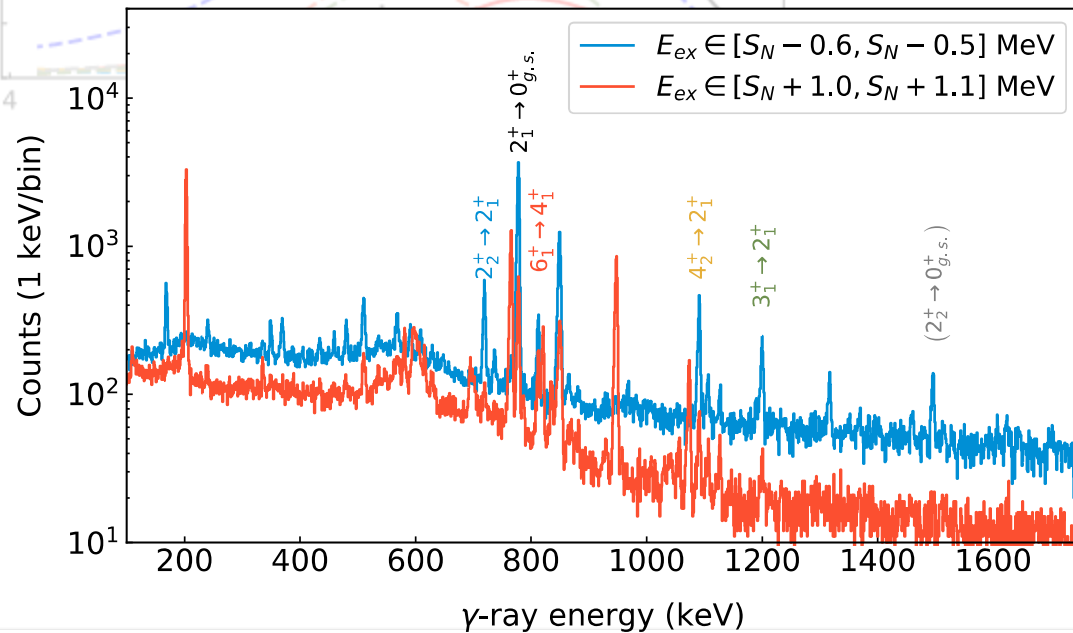


- **Absorption** of the neutron as a function of excitation **energy** and **spin** computed with **GFT** formalism
- We used the phenomenological **Koning-Delaroche OP**

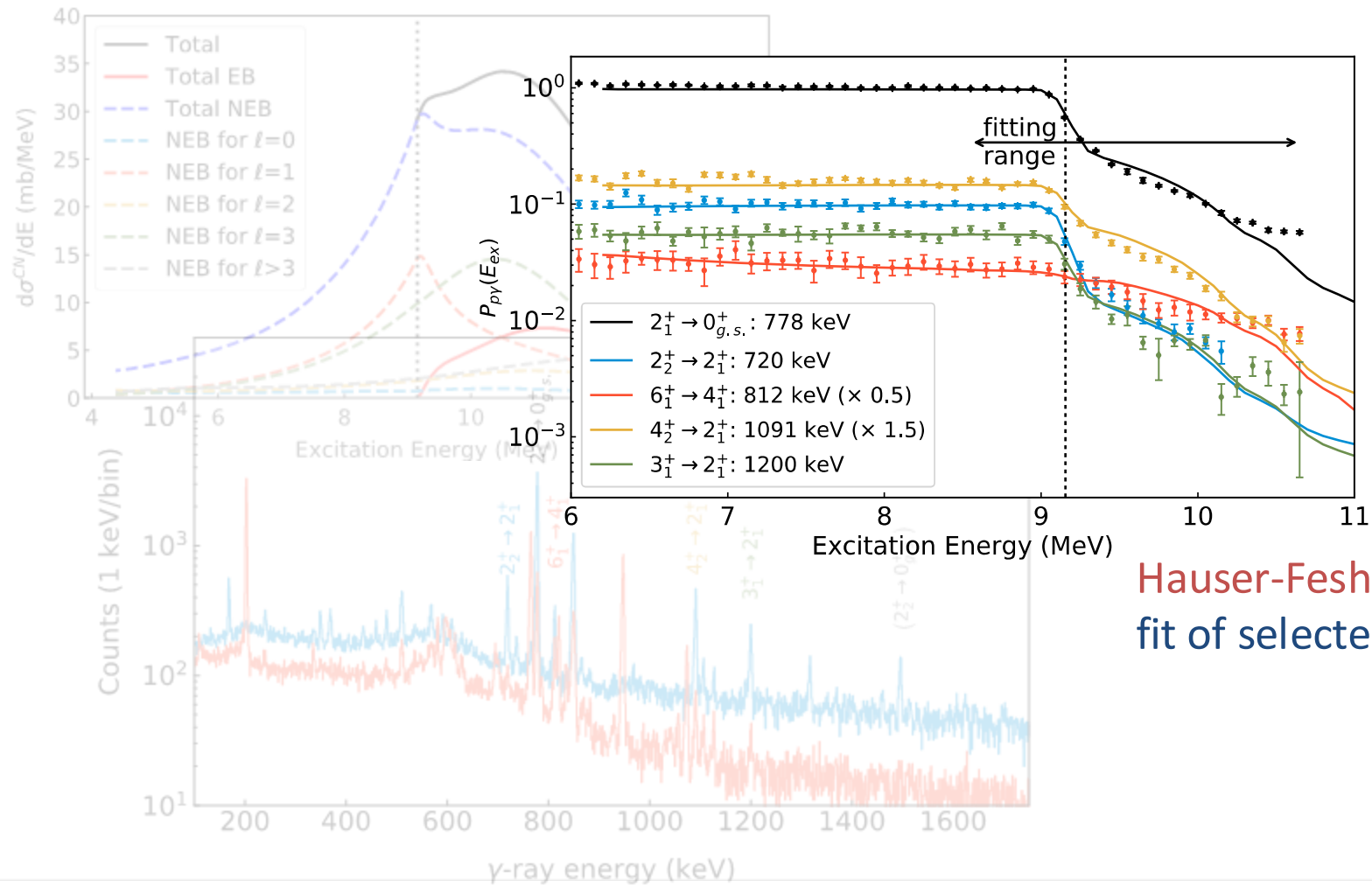
GFT applied to the Surrogate Method: $^{95}\text{Mo}(\text{d},\text{p}\gamma)$ with Koning-Delaroche OP



- γ rays observed in coincidence with protons
- transitions from both ^{95}Mo and ^{96}Mo are identified



GFT applied to the Surrogate Method: $^{95}\text{Mo}(\text{d},\text{p}\gamma)$ with **Koning-Delaroche** OP



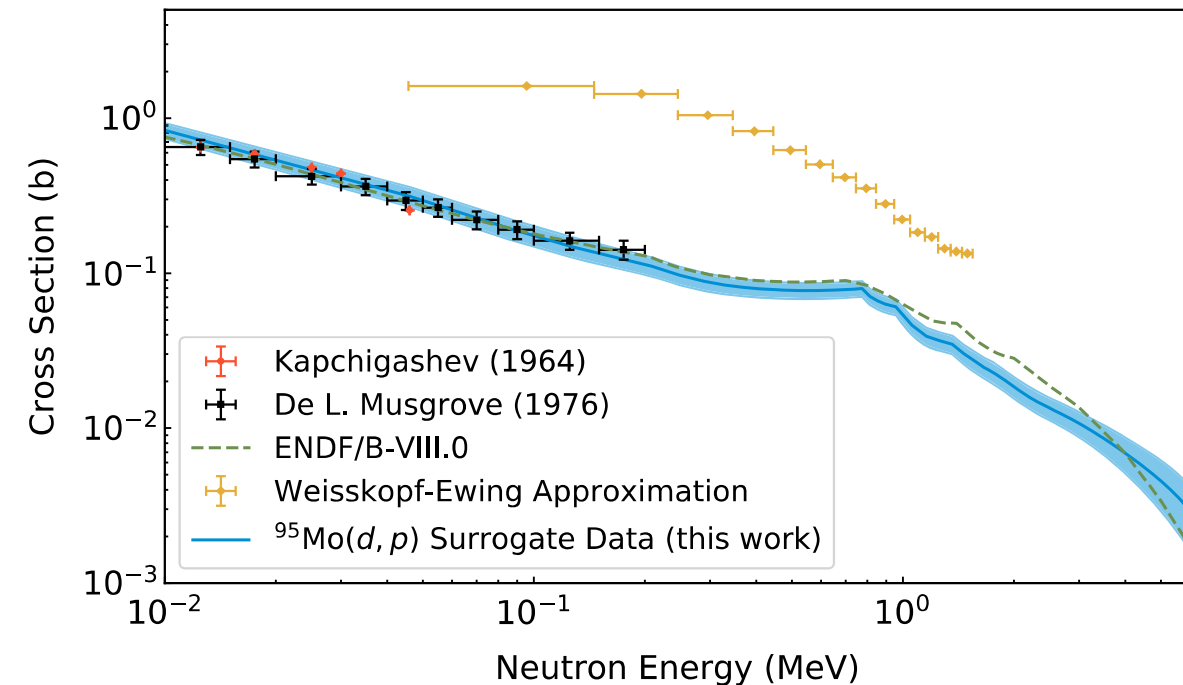
Hauser-Feshbach parameters determined by
fit of selected γ lines (J. Escher)

GFT applied to the Surrogate Method: $^{95}\text{Mo}(d,p\gamma)$ with Koning-Delaroche OP

PHYSICAL REVIEW LETTERS **122**, 052502 (2019)

Towards Neutron Capture on Exotic Nuclei: Demonstrating $(d,p\gamma)$ as a Surrogate Reaction for (n,γ)

A. Ratkiewicz,^{1,2,*} J. A. Cizewski,² J. E. Escher,¹ G. Potel,^{3,4} J. T. Burke,¹ R. J. Casperson,¹
M. McCleskey,⁵ R. A. E. Austin,⁶ S. Burcher,² R. O. Hughes,^{1,7} B. Manning,² S. D. Pain,⁸
W. A. Peters,⁹ S. Rice,² T. J. Ross,⁷ N. D. Scielzo,¹ C. Shand,^{2,10} and K. Smith¹¹



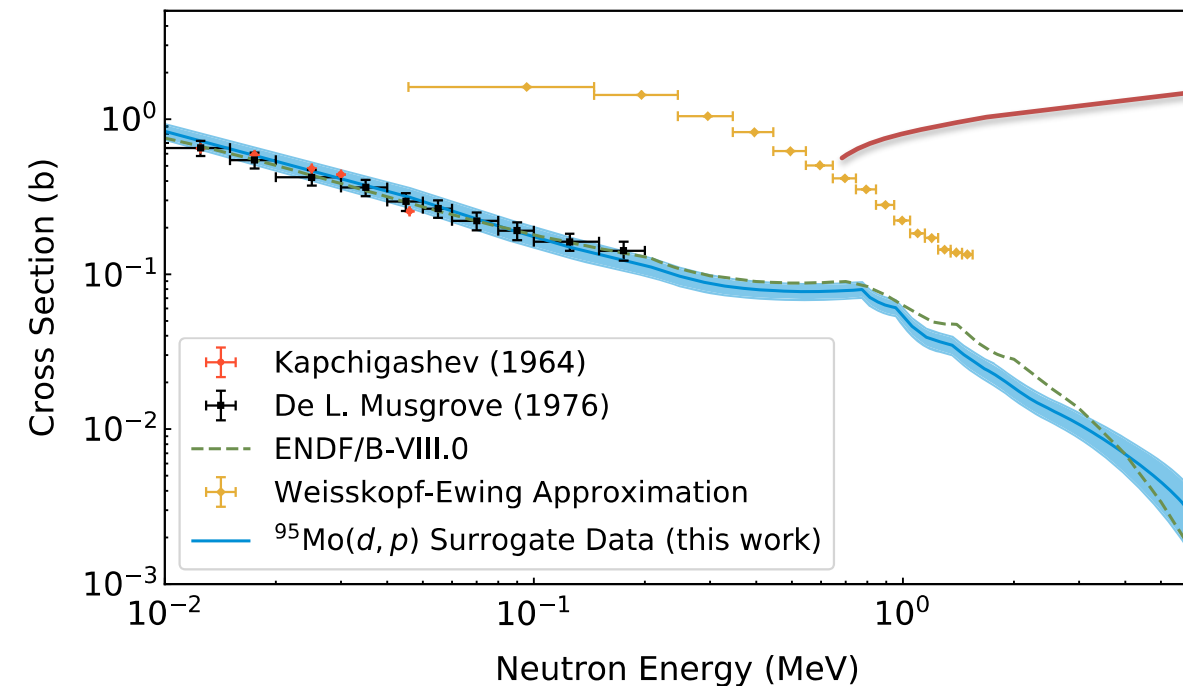
- The obtained Hauser-Feshbach parameters are used to calculate (n,γ)
- We found an excellent agreement with the direct measurement.

GFT applied to the Surrogate Method: $^{95}\text{Mo}(d,p\gamma)$ with Koning-Delaroche OP

PHYSICAL REVIEW LETTERS **122**, 052502 (2019)

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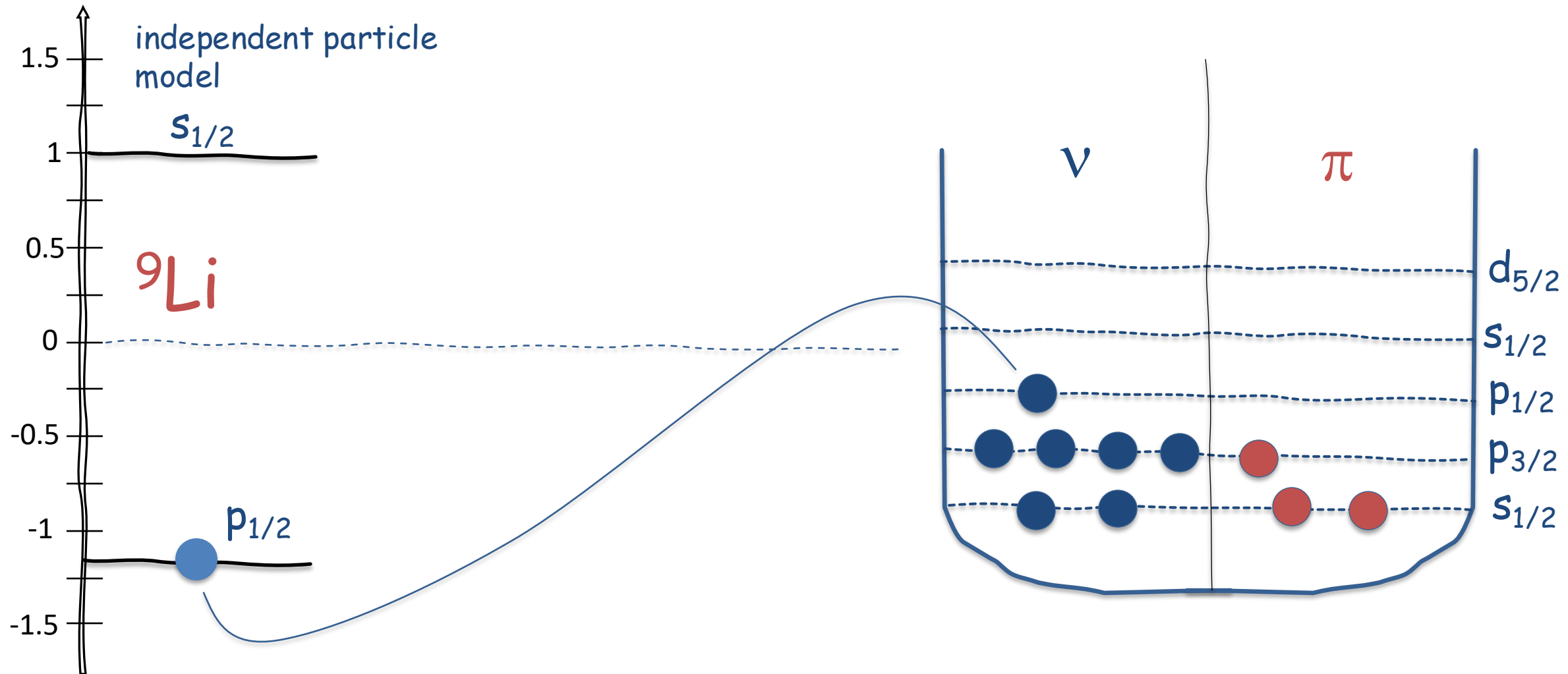
A. Ratkiewicz,^{1,2,*} J. A. Cizewski,² J. E. Escher,¹ G. Potel,^{3,4} J. T. Burke,¹ R. J. Casperson,¹
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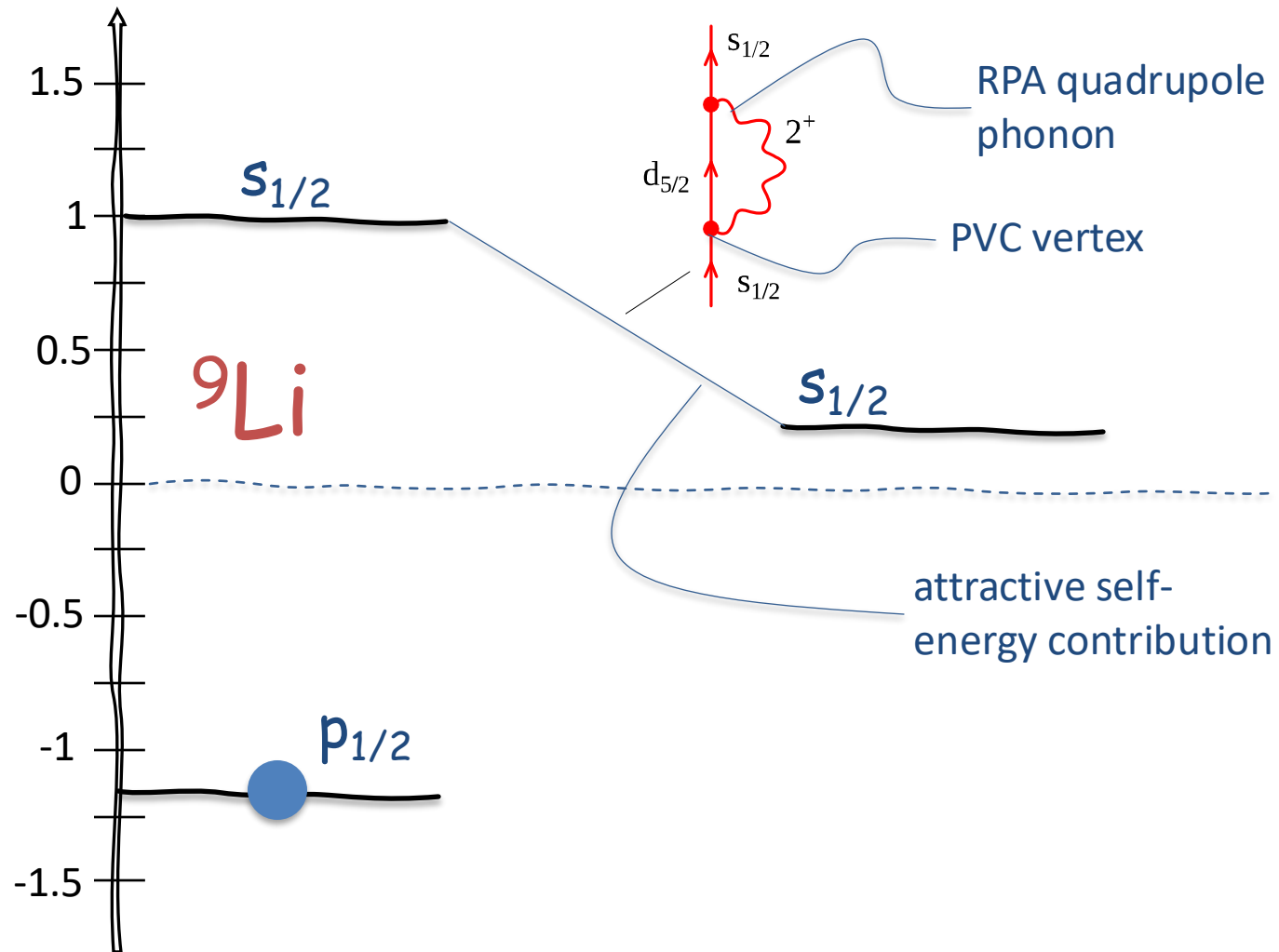
a failure to account for the initial spin distribution (Weisskopf-Ewing approximation) leads to poor results!

- The obtained Hauser-Feshbach parameters are used to calculate (n,γ)
- We found an excellent agreement with the direct measurement.

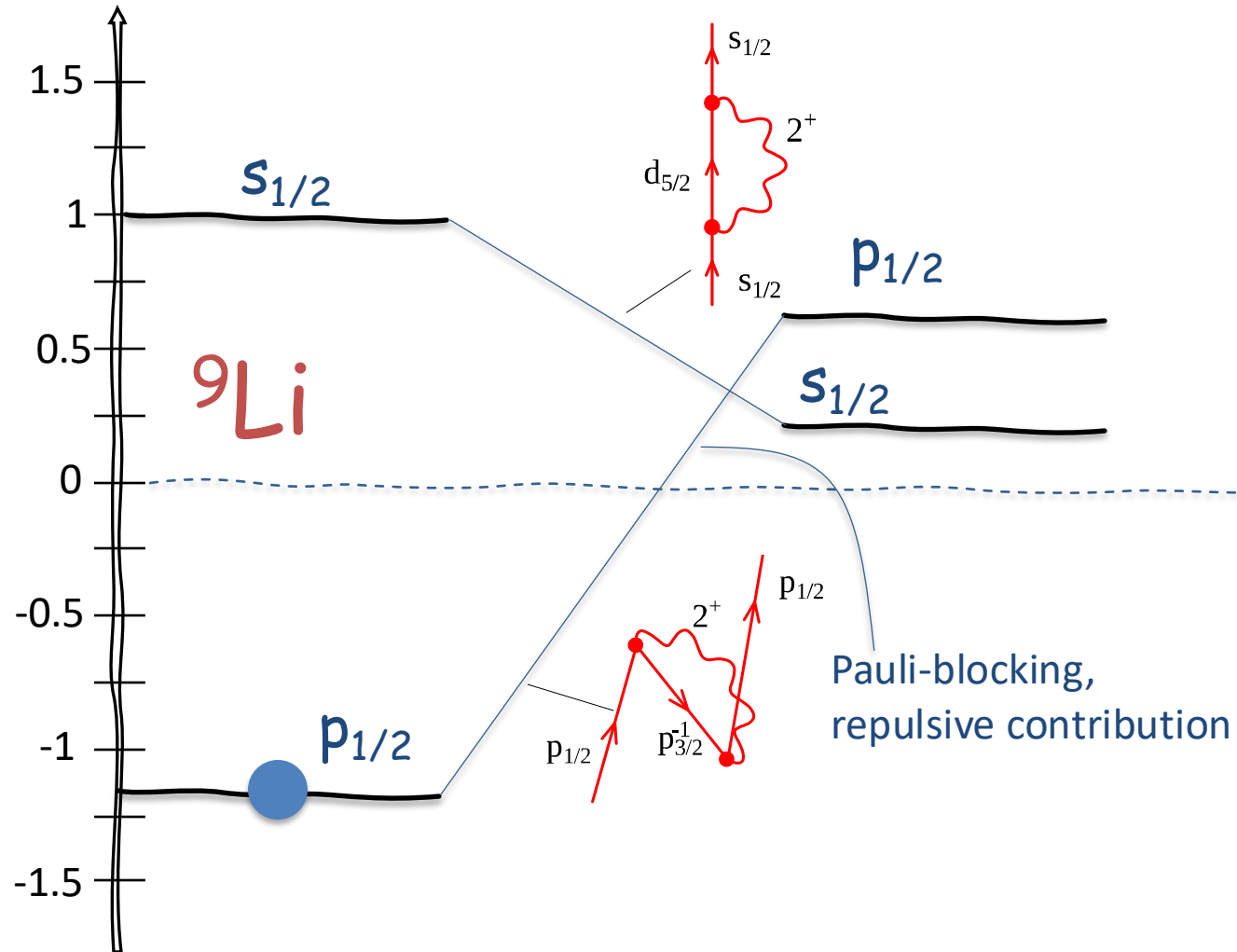
GFT with Nuclear Field Theory (NFT): ${}^9\text{Li}(d,p)$



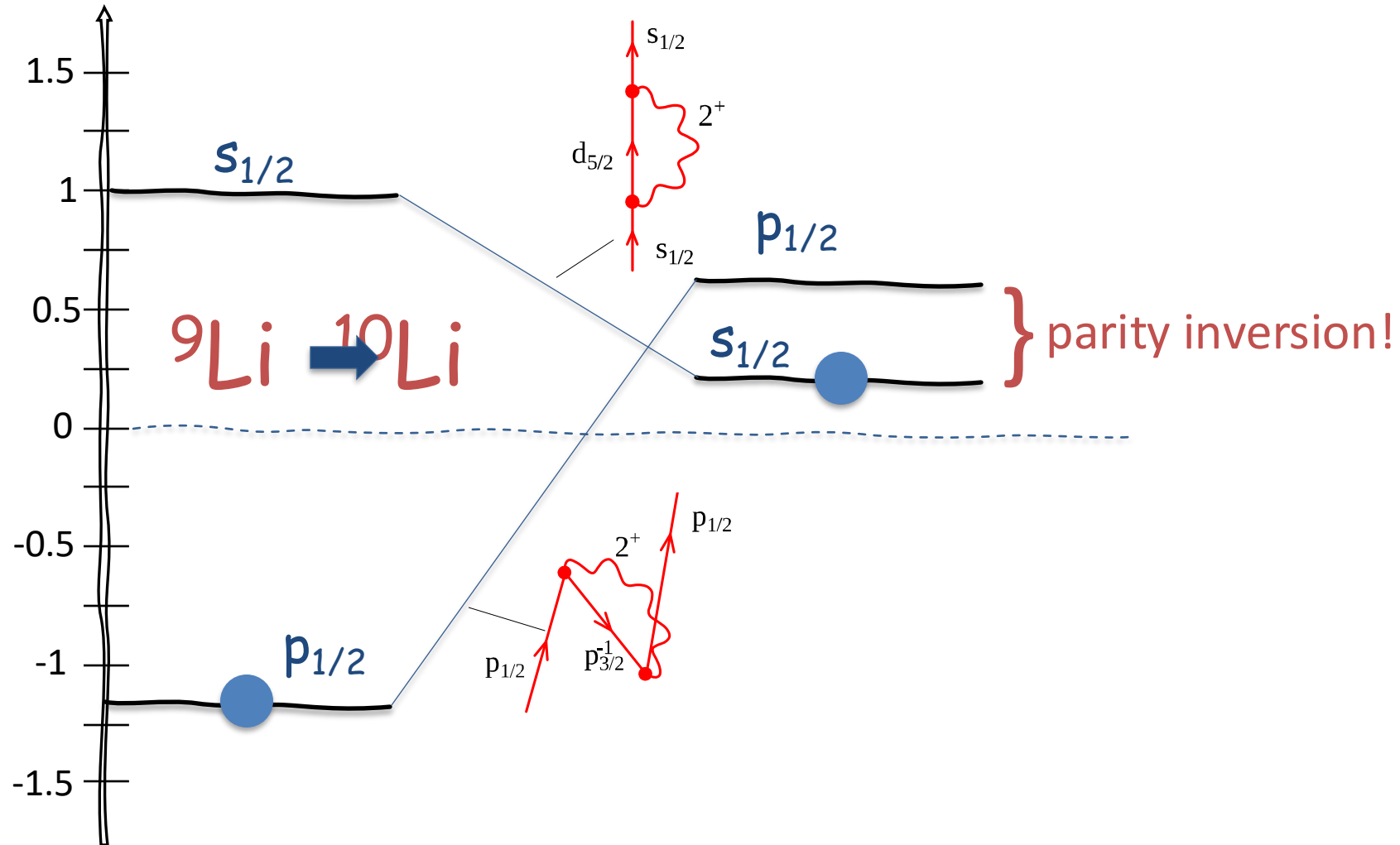
GFT with Nuclear Field Theory (NFT): ${}^9\text{Li}(d,p)$



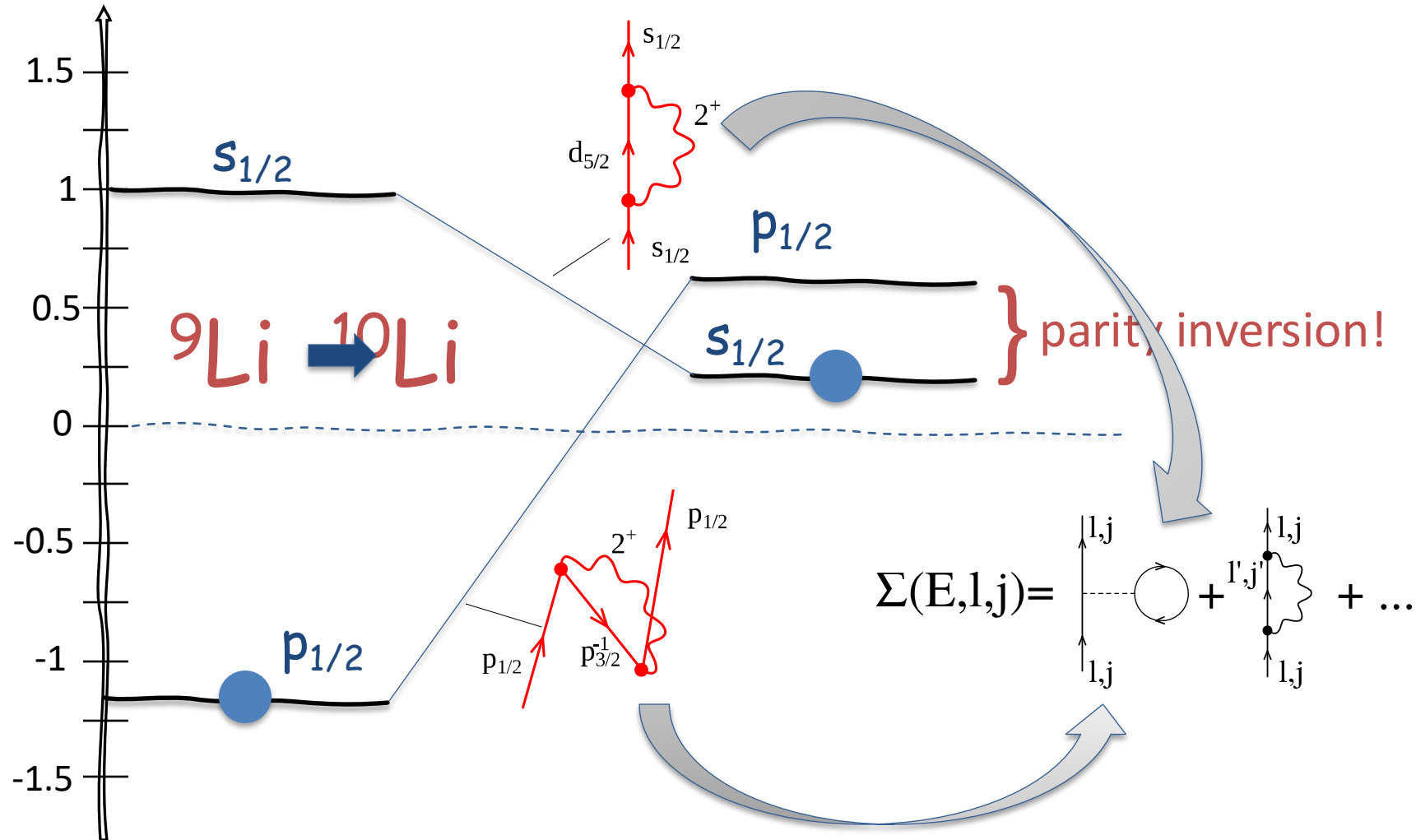
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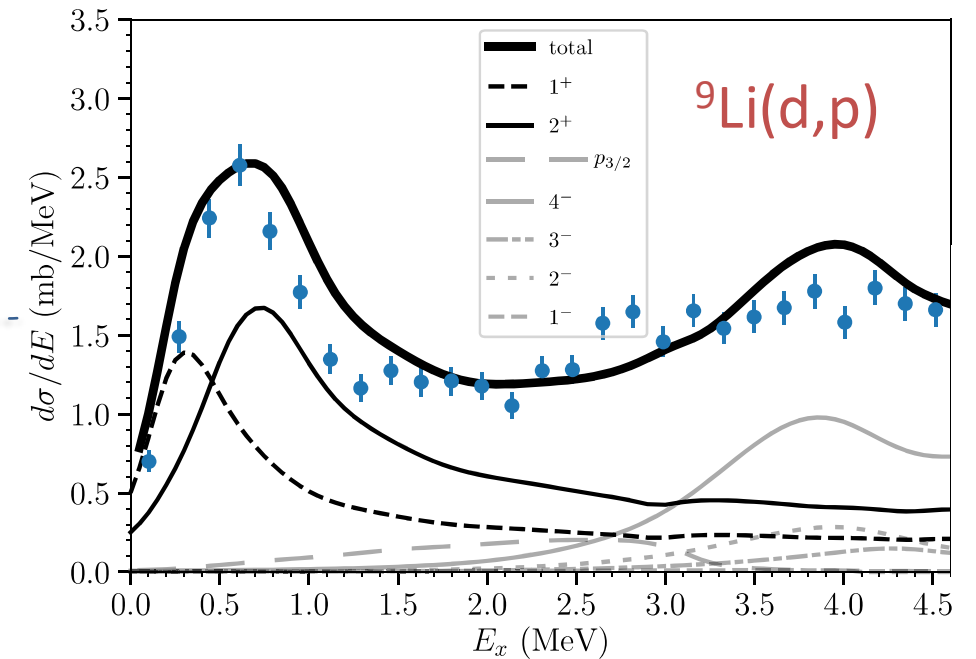
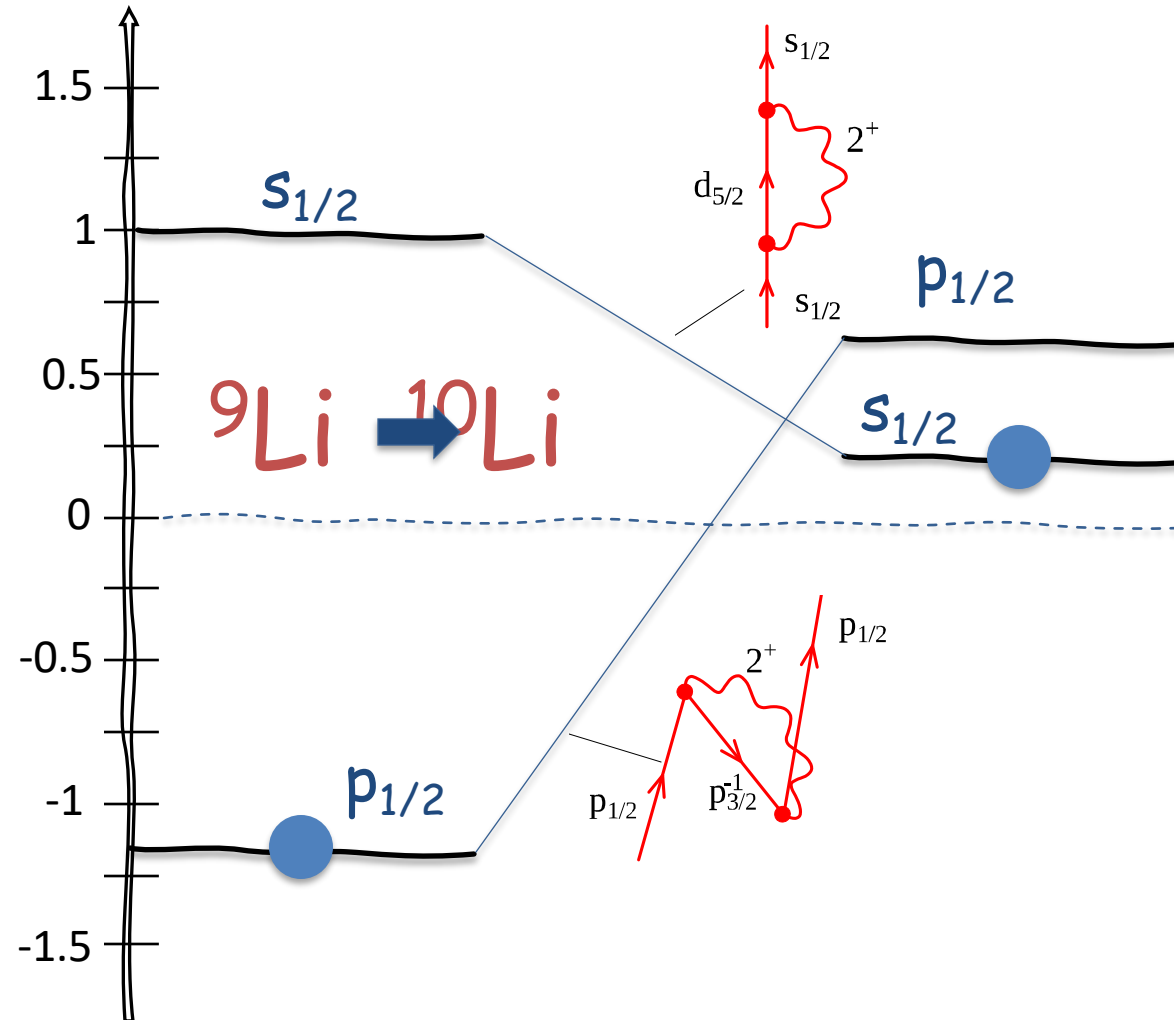
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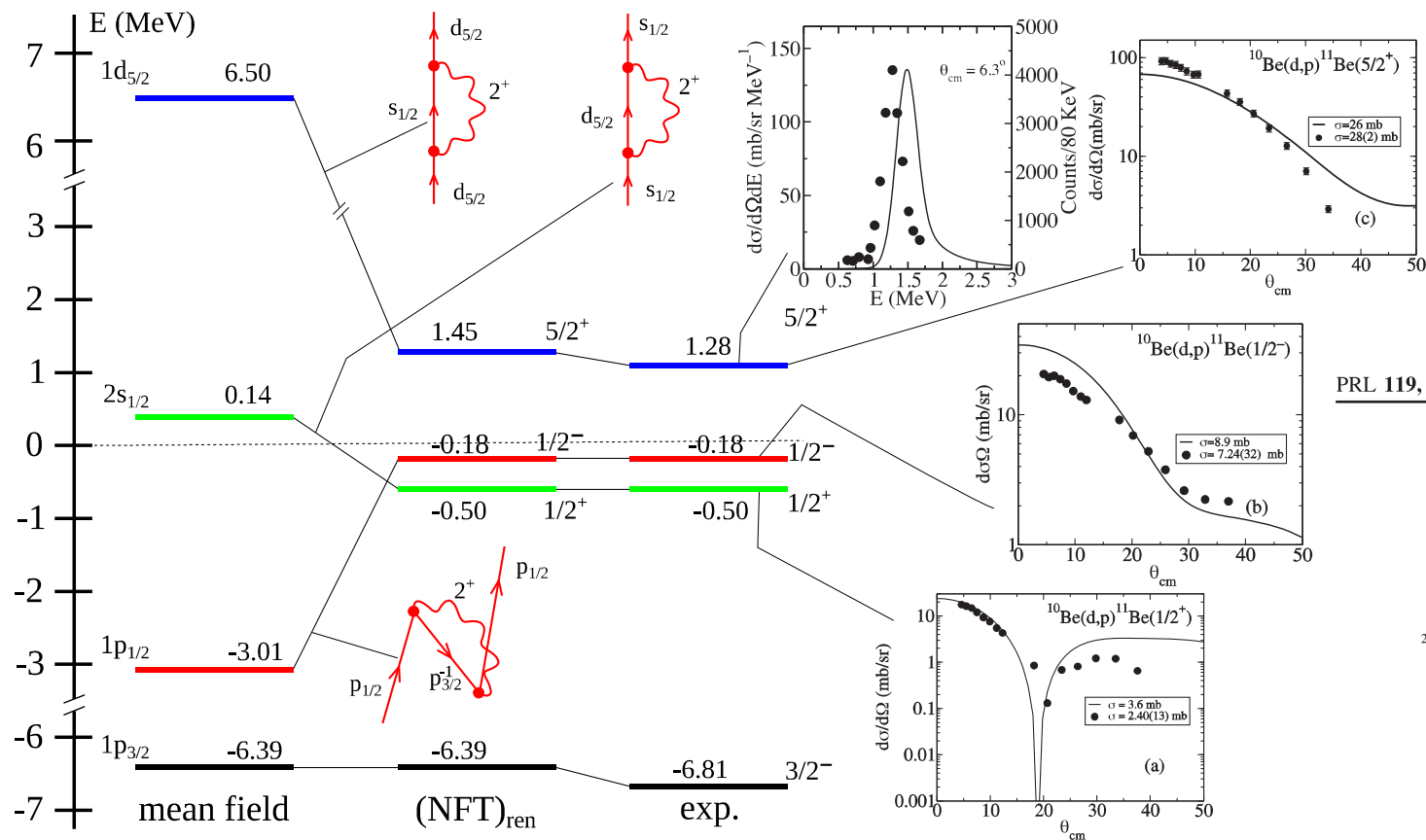
Cavallaro *et al.*, PRL **118**, 012701 (2017)

Barranco, GP, Vigezzi, Broglia PRC **101**, 031305(R) (2020)



theoretical description validated by experiment

It works for ^{11}Be , too!



PRL 119, 082501 (2017)

PHYSICAL REVIEW LETTERS

week ending
25 AUGUST 2017

Structure and Reactions of ^{11}Be : Many-Body Basis for Single-Neutron Halo

F. Barranco,¹ G. Potel,² R. A. Broglia,^{3,4} and E. Vigezzi⁵

¹Departamento de Física Aplicada III, Escuela Superior de Ingenieros, Universidad de Sevilla, Camino de los Descubrimientos, 41092 Sevilla, Spain

²National Superconducting Cyclotron Laboratory, Michigan State University, East Lansing, Michigan 48824, USA

³The Niels Bohr Institute, University of Copenhagen, DK-2100 Copenhagen, Denmark

⁴Dipartimento di Fisica, Università degli Studi Milano, Via Celoria 16, I-20133 Milano, Italy

⁵INFN Sezione di Milano, Via Celoria 16, I-20133 Milano, Italy

Barranco, GP, Broglia, Vigezzi PRL 119, 082501 (2017)

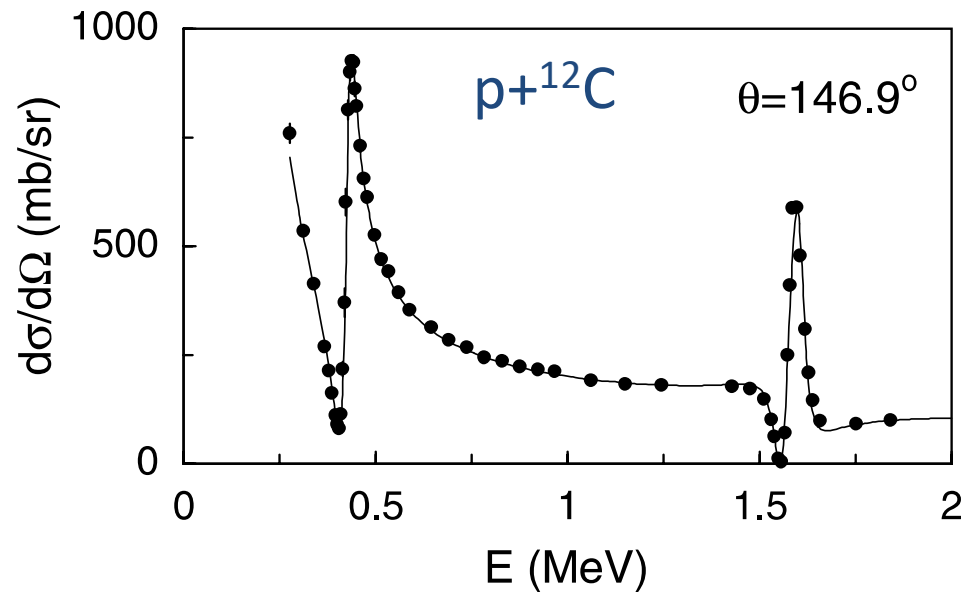
Part 5

Teaser: connections with R-Matrix and Hauser-Feshbach

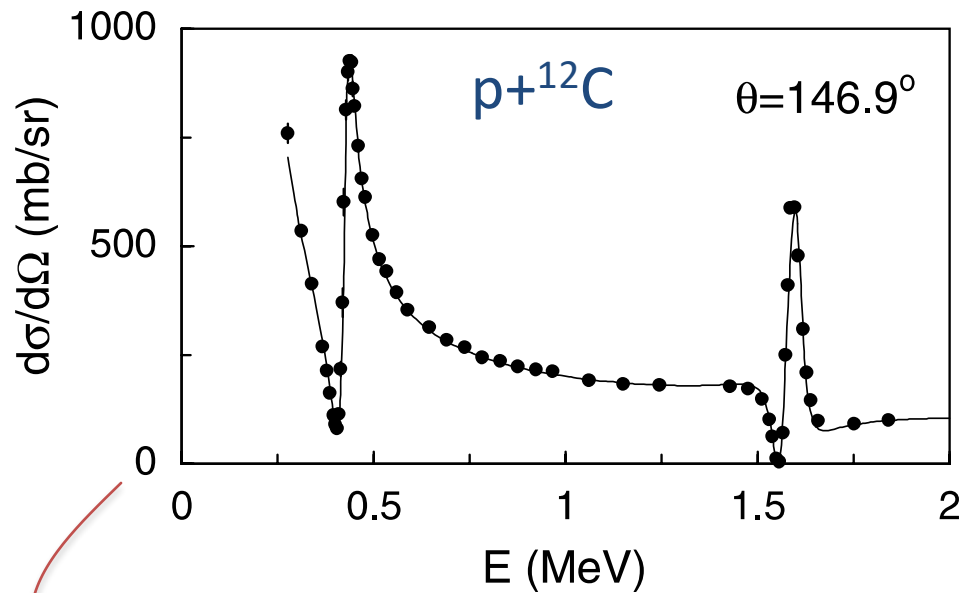


with C. Hebborn

The GFT formalism suggests an *R*-matrix parametrization for the indirect cross section



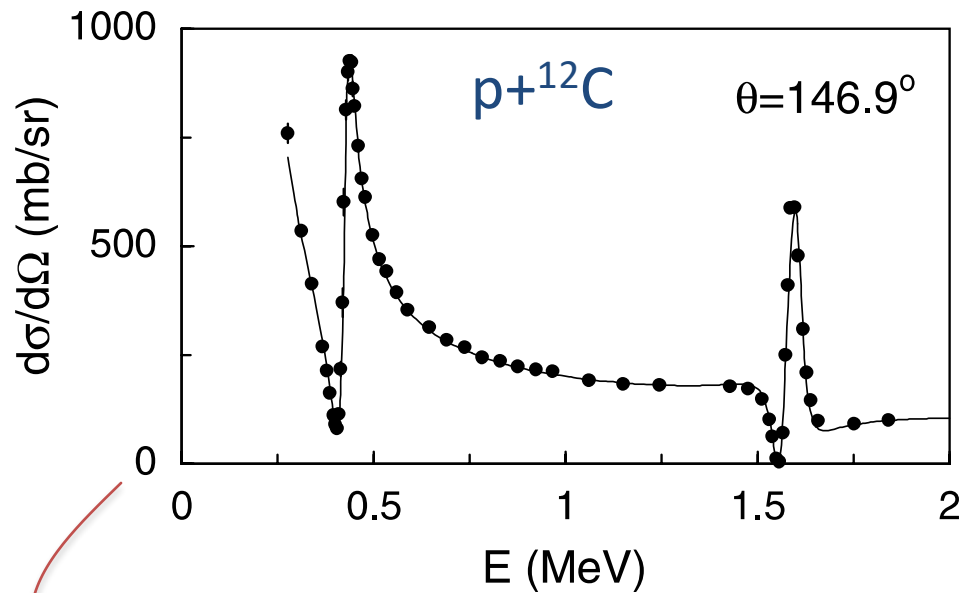
The GFT formalism suggests an R -matrix parametrization for the indirect cross section



$$T_{i0}(E) = \sqrt{P_i(E)P_0(E)} \sum_{pq} \frac{\gamma_{ip}\gamma_{0q}}{(E_p - E)\delta_{pq} - \sum_c \gamma_{ic}\gamma_{jc}(S_c(E) + iP_c(E))}$$

T-matrix partial widths and energy parameters fitted from data

The GFT formalism suggests an *R*-matrix parametrization for the indirect cross section



$$\frac{d\sigma(E)}{d\sigma} \propto |T_{i0}(E)|^2$$

$$T_{i0}(E) = \sqrt{P_i(E)P_0(E)} \sum_{pq} \frac{\gamma_{ip}\gamma_{0q}}{(E_p - E)\delta_{pq} - \sum_c \gamma_{ic}\gamma_{jc}(S_c(E) + iP_c(E))}$$

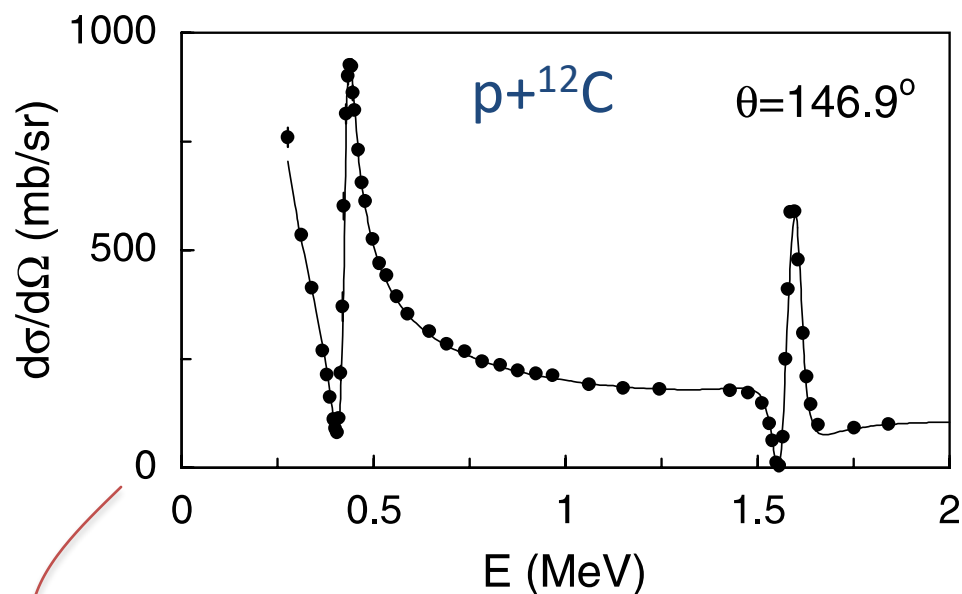
T-matrix partial widths and energy parameters fitted from data

The GFT formalism suggests an *R*-matrix parametrization for the indirect cross section

Rep. Prog. Phys. **73** (2010) 036301 (44pp)

The *R*-matrix theory

P Descouvemont¹ and D Baye^{1,2}



$$\frac{d\sigma(E)}{d\sigma} \propto |T_{i0}(E)|^2$$

$$T_{i0}(E) = \sqrt{P_i(E)P_0(E)} \sum_{pq} \frac{\gamma_{ip}\gamma_{0q}}{(E_p - E)\delta_{pq} - \sum_c \gamma_{ic}\gamma_{jc}(S_c(E) + iP_c(E))}$$

T-matrix partial widths and energy parameters fitted from data

The GFT formalism suggests an R -matrix parametrization for the indirect cross section

connection between direct and indirect R -matrix parameters

example:

- direct: α scattering ($T_{i0}(E)$)
- indirect: (${}^6\text{Li}, d$). ($T'_{i0}(E)$)

The GFT formalism suggests an R -matrix parametrization for the indirect cross section

connection between direct and indirect R -matrix parameters
example:

- direct: α scattering ($T_{i0}(E)$)
- indirect: (${}^6\text{Li}, d$). ($T_{i0}^I(E)$)

indirect T-
matrix

$$T_{i0}^I(E) = \int T_{i0}(E_k) g(\mathbf{k}; E) d\mathbf{k}.$$

direct T-
matrix

The GFT formalism suggests an R -matrix parametrization for the indirect cross section

connection between direct and indirect R -matrix parameters
example:

- direct: α scattering ($T_{i0}(E)$)
- indirect: (${}^6\text{Li}, d$). ($T_{i0}^I(E)$)

indirect T-matrix

$$T_{i0}^I(E) = \int T_{i0}(E_k) g(\mathbf{k}; E) d\mathbf{k}.$$

direct T-matrix

$$E_k = \frac{\hbar^2 k^2}{2\mu}$$

The GFT formalism suggests an R -matrix parametrization for the indirect cross section

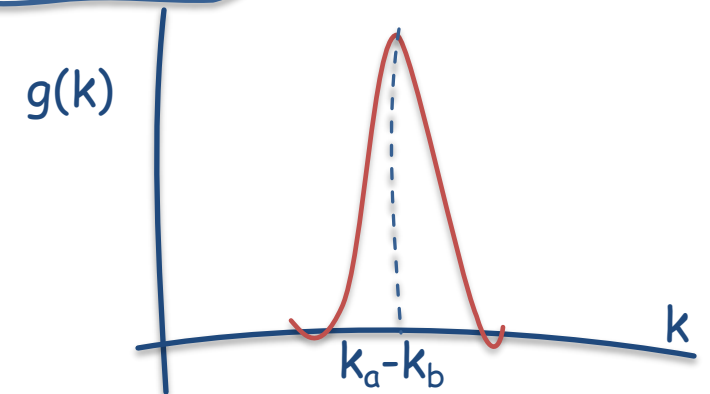
connection between direct and indirect R -matrix parameters
example:

- direct: α scattering ($T_{i0}(E)$)
- indirect: (${}^6\text{Li}, d$). ($T_{i0}^I(E)$)

indirect T-
matrix

$$T_{i0}^I(E) = \int T_{i0}(E_k) g(\mathbf{k}; E) d\mathbf{k}.$$

direct T-
matrix



broadening
factor

$$g(\mathbf{k}; E) = \int \psi^{HM}(\mathbf{r}_{xA}; E) F^*(\mathbf{r}_{xA}, \mathbf{k}) d\mathbf{r}_{xA}$$

The GFT formalism suggests an R -matrix parametrization for the indirect cross section

connection between direct and indirect R -matrix parameters

example:

- direct: α scattering
- indirect: (${}^6\text{Li}, d$)

indirect T-matrix

$$T_{i0}^I = \int \sqrt{P_i(E_k)P_0(E_k)} \sum_{pq} \frac{\gamma_{ip}\gamma_{0q}}{(E_p - E_k)\delta_{pq} - \sum_c \gamma_{ic}\gamma_{jc}(S_c(E_k) + iP_c(E_k))} g(\mathbf{k}) d\mathbf{k}.$$

$$E_k = \frac{\hbar^2 k^2}{2\mu}$$

broadening factor

$$g(\mathbf{k}) = \int \psi^{HM}(\mathbf{r}_{xA}) F^*(\mathbf{r}_{xA}, \mathbf{k}) d\mathbf{r}_{xA}$$

The GFT formalism suggests an R -matrix parametrization for the indirect cross section

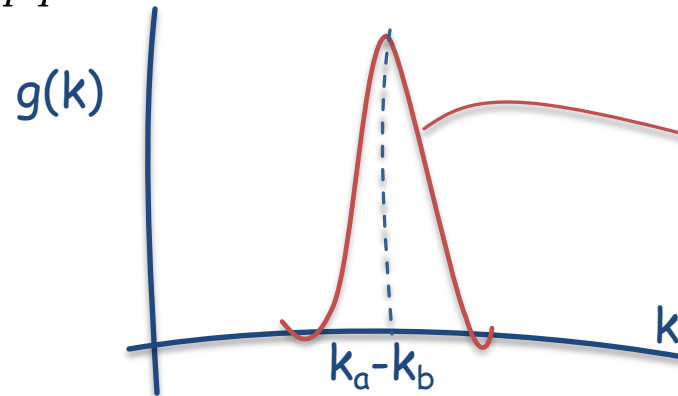
connection between direct and indirect R -matrix parameters

example:

- direct: α scattering
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indirect T-
matrix

$$T_{i0}^I = \int \sqrt{P_i(E_k)P_0(E_k)} \sum_{pq} \frac{\gamma_{ip}\gamma_{0q}}{(E_p - E_k)\delta_{pq} - \sum_c \gamma_{ic}\gamma_{jc}(S_c(E_k) + iP_c(E_k))} g(\mathbf{k}) d\mathbf{k}.$$



$$g(\mathbf{k}) = \int \psi^{HM}(\mathbf{r}_{xA}) F^*(\mathbf{r}_{xA}, \mathbf{k}) d\mathbf{r}_{xA}$$

The GFT formalism suggests an *R*-matrix parametrization for the indirect cross section

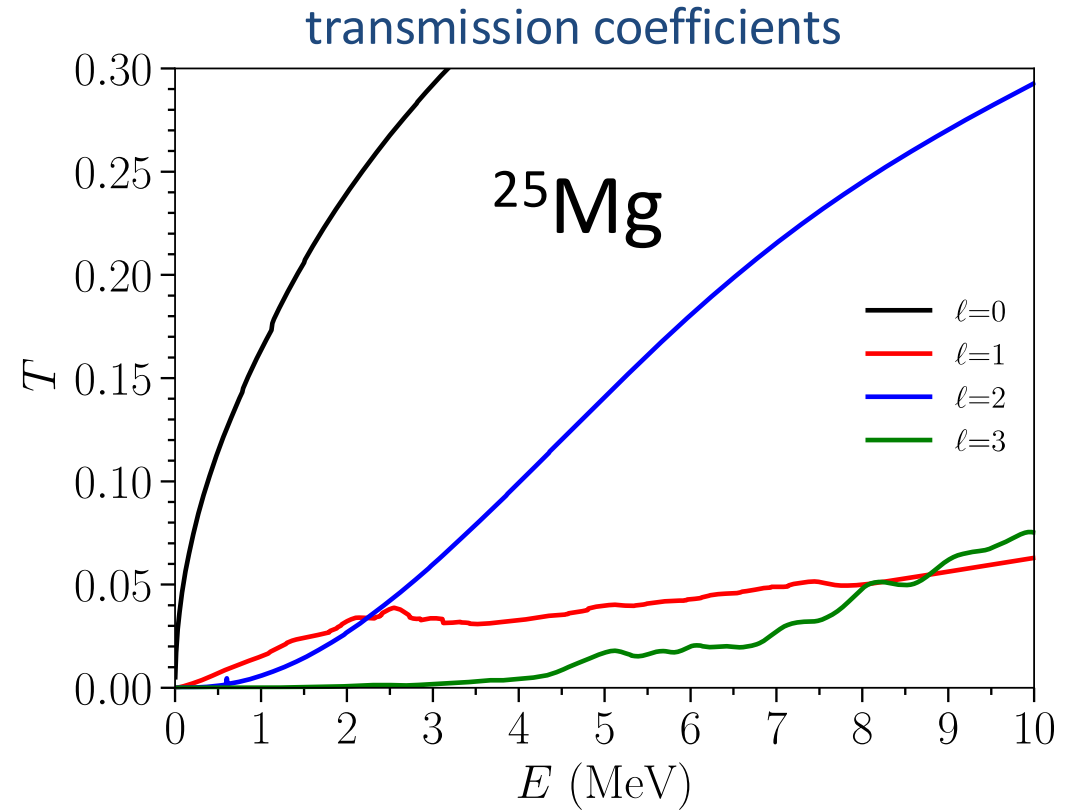
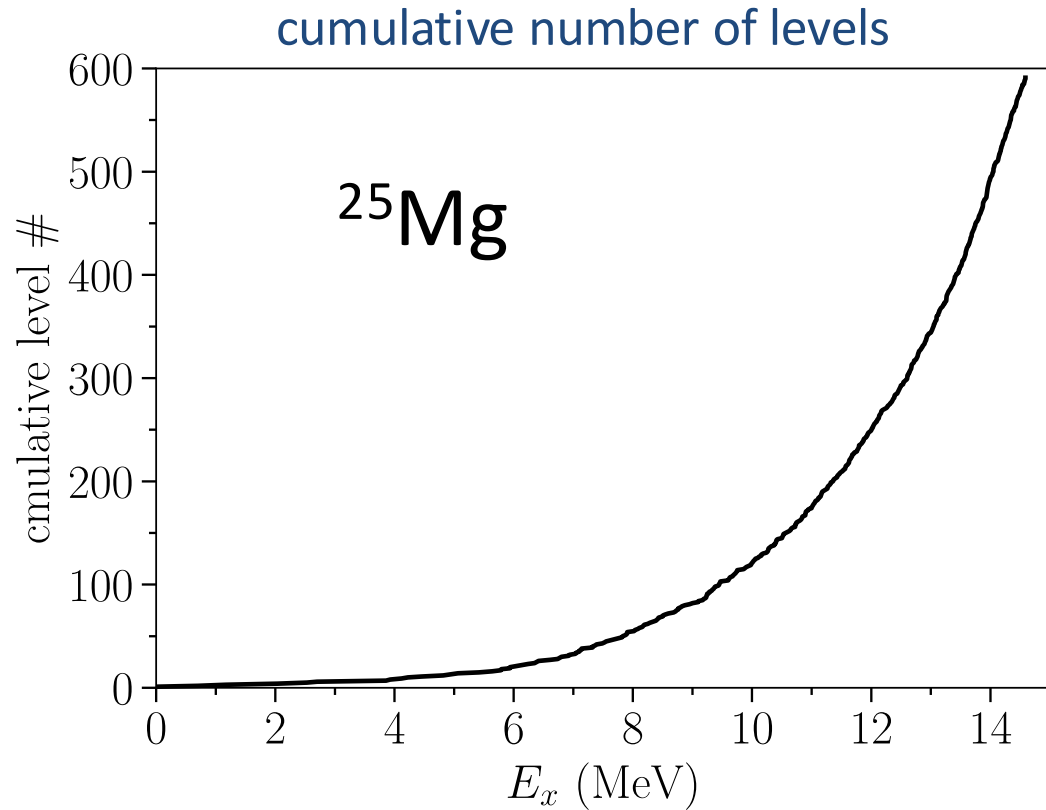
- If the broadening distribution is narrow, the T-matrix can be evaluated at the peak
- This is essentially the approximation made by Barker in *Aust. J. Phys.* **20** (341) 1967 for isolated resonances

$$T_{i0}^I = \int \sqrt{P_i(E_k)P_0(E_k)} \sum_{pq} \frac{\gamma_{ip}\gamma_{0q}}{(E_p - E_k)\delta_{pq} - \sum_c \gamma_{ic}\gamma_{jc}(S_c(E_k) + iP_c(E_k))} g(\mathbf{k}) d\mathbf{k}.$$



$$T_{i0}^I \approx \sqrt{P_i(E_k^{max})P_0(E_k^{max})} \sum_{pq} \frac{\gamma_{ip}\gamma_{0q}}{(E_p - E_k^{max})\delta_{pq} - \sum_c \gamma_{ic}\gamma_{jc}(S_c(E_k^{max}) + iP_c(E_k^{max}))} \int g(\mathbf{k}) d\mathbf{k}.$$

The OP, the level density, and the γ strength function are connected through the same underlying physics



We can explicitly connect with statistical model (Hauser-Feshbach approach) through energy-averaging

Conclusions and some perspectives

- Phenomenology has to be complemented with **theory** for reliable extrapolation **across different regimes**.
- The calculation of the **OP** provides a flexible and **versatile** path, including **3-body** reactions (with **GFT**).

what's next?

- Improve **microscopic** inputs.
- Disentangle **direct, pre-equilibrium, and compound** reactions.
- Explore **the limits of validity** of the statistical model.
- Symmetry-breaking systems (**deformation, pairing**).
- **R-Matrix** parametrization of **indirect** reactions.

Collaborators:

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- E. Vigezzi (INFN, Milano)
- F. Barranco (University of Seville)
- **G. Sargsyan**, F. Nunes (MSU)
- C. Hebborn (IJCLab, Paris)

⁹⁵**Mo:** A. Ratkiewicz, J. Escher, J. Burke, R. Casperson, R. Hughes, N. Scielzo (LLNL), J. Cizewski, S. Burcher, B. Manning, S. Rice, C. Shand (Rutgers), M. McCleskey (TAMU), R. Austin (St Mary's), S. Pain (ORNL), W. Peters (U of Tennessee), T. Ross (U of Richmond) and K. Smith (LANL).

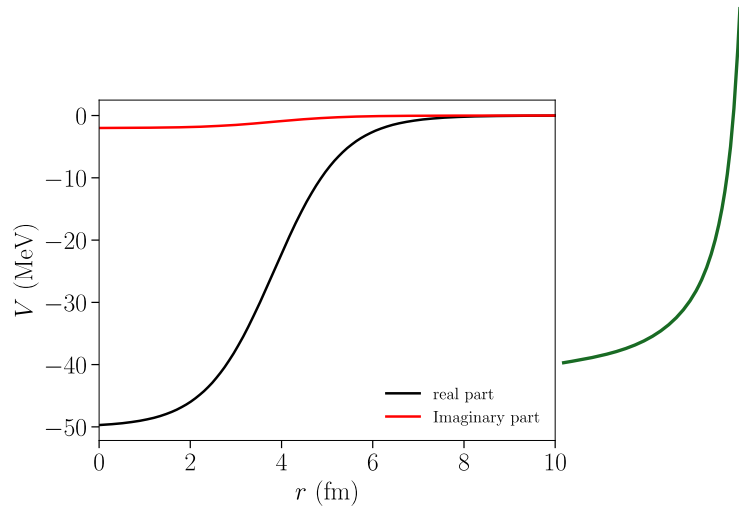
Thank you!



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^{40}Ca OP calculated in a weak coupling, collective model approximation

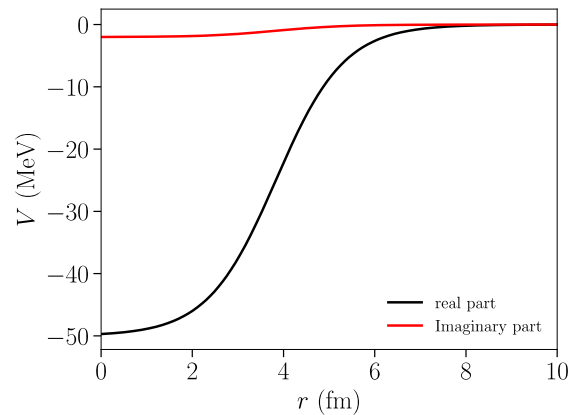
$$V(\mathbf{r}, \mathbf{r}'; E) = U_0(r) + \sum_i U_{0i}(\mathbf{r}) G(E - E_i, \mathbf{r}, \mathbf{r}') U_{i0}(\mathbf{r}')$$



- the **static** potential is a simple **Woods-Saxon**
- a small imaginary part **W** is included to account for the **lack of absorption** of the model
- this is a consequence of the **over-simplification** of the spectrum
- The small imaginary part **spoils** dispersivity

^{40}Ca OP calculated in a weak coupling, collective model approximation

$$V(\mathbf{r}, \mathbf{r}'; E) = U_0(r) + \sum_i U_{0i}(\mathbf{r}) G(E - E_i, \mathbf{r}, \mathbf{r}') U_{i0}(\mathbf{r}')$$



- the spectrum of ^{40}Ca is **approximated** by 6 collective **vibrational** states
- the **deformation** parameters β_λ are constrained by the experimental **inelastic** scattering cross section

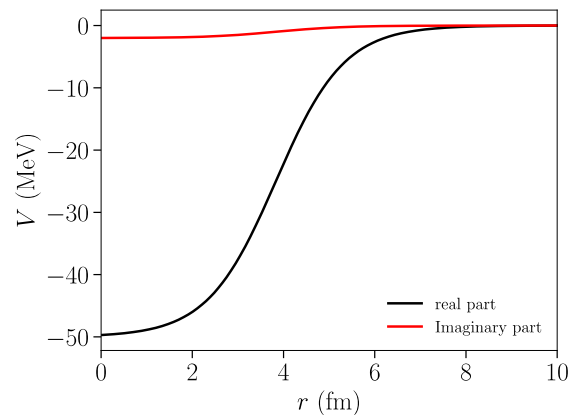
$$|\Phi(^{41}\text{Ca})\rangle_i \approx |\Phi(^{40}\text{Ca})\rangle_i \otimes |\chi(n)\rangle_i$$

λ^π	2^+	3^-	4^+	5^-	6^+	7^-	8^+
σ_A (mb)	176	150	116	106	87	80	68

that the isovector coupling strength is 0.6 times the isoscalar $^{20}\lambda$. The observed
From Rao, Reeves, and Satchler, NPA **207** (1973) 182

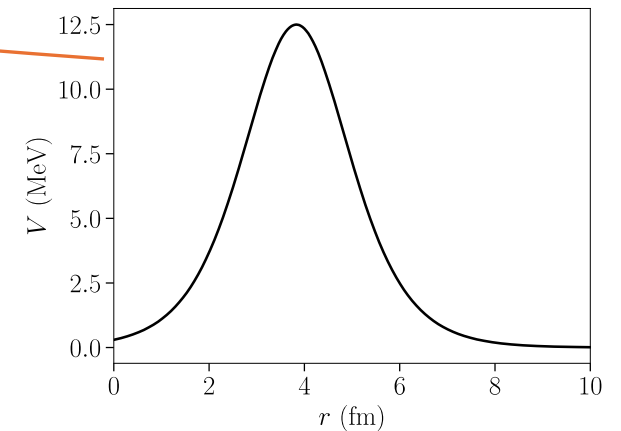
^{40}Ca OP calculated in a weak coupling, collective model approximation

$$V(\mathbf{r}, \mathbf{r}'; E) = U_0(r) + \sum_i U_{0i}(\mathbf{r}) G(E - E_i, \mathbf{r}, \mathbf{r}') U_{i0}(\mathbf{r}')$$



- the couplings are **surface** peaked
- by construction, the **experimental** contribution of each vibrational state to the absorption cross section is

reproduced

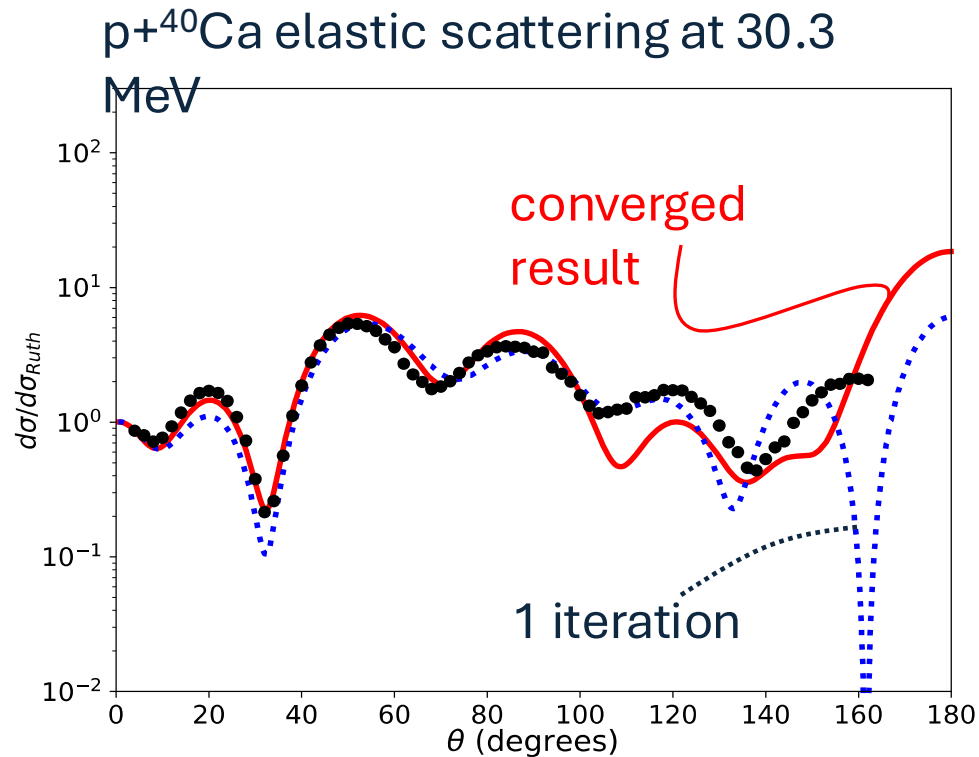


λ^π	2^+	3^-	4^+	5^-	6^+	7^-	8^+
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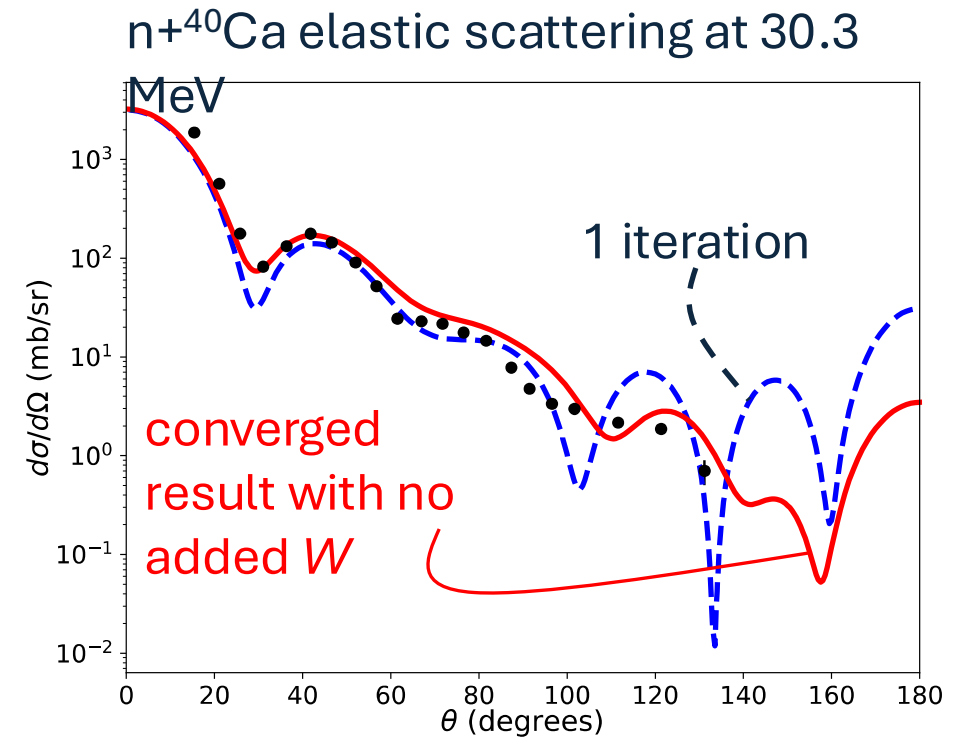
$$U_{0i}(\mathbf{r}) \sim \beta_i \frac{dU(r)}{dr} Y^{\lambda_i}(\hat{r})$$

that the isovector coupling strength is 0.6 times the isoscalar ²⁰⁾ The observed
From Rao, Reeves, and Satchler, NPA **207** (1973) 182

We benchmark our results against Rao *et al.*, and look at the effect of iterations



- 1 iteration calculation agrees with Rao *et al.* (not shown)
- Converged result different at large angles



- Good result for neutrons just by removing Coulomb
- Added non-dispersive imaginary part W not needed for the converged result

ab initio methods applied to reactions (NCSMC)

PHYSICAL REVIEW LETTERS **129**, 042503 (2022)

Ab Initio Prediction of the $^4\text{He}(d,\gamma)^6\text{Li}$ Big Bang Radiative Capture

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$$|\Psi^{J^{\pi}T}\rangle = \sum_{\lambda} c_{\lambda}^{J^{\pi}T} |A\lambda J^{\pi}T\rangle + \sum_{\nu} \int_0^{+\infty} dr r^2 \frac{\gamma_{\nu}^{J^{\pi}T}(r)}{r} \hat{A}_{\nu} |\Phi_{\nu r}^{J^{\pi}T}\rangle$$

continuum scattering wf

clusters in NCSM

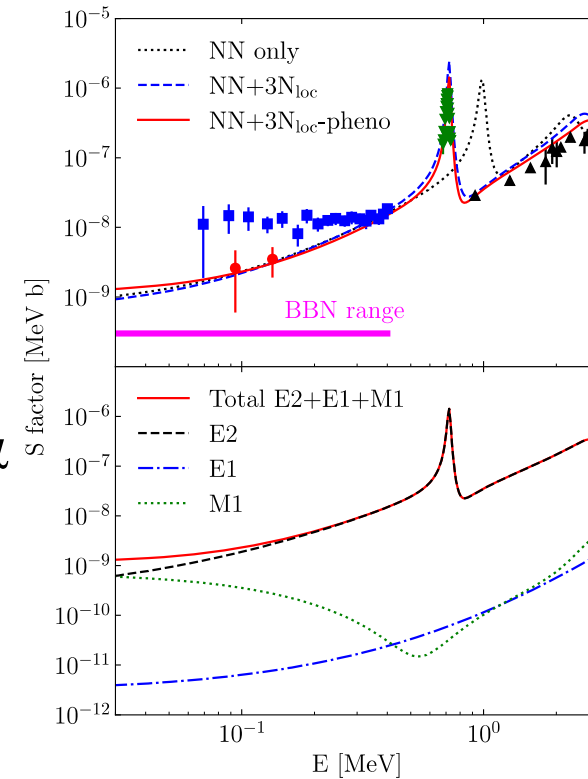
NCSMC wf used in T-matrix

$$\sigma(E) = \frac{64\pi^4}{4\pi\epsilon_0\hbar v} \sum_{\kappa\lambda} \frac{k_{\gamma}^{2\lambda+1}}{[(2\lambda+1)!!]^2} \frac{\lambda+1}{\lambda}$$

$$\times \sum_{J_i l_i s_i} \frac{\hat{J}_f^2}{\hat{s}_p^2 \hat{s}_T^2 \hat{l}_i^2} |\langle \Psi_{J_f}^{J^{\pi}T_f} || \mathcal{M}^{\kappa\lambda} || \Psi_{l_i s_i}^{J^{\pi}T_i} \rangle|^2$$

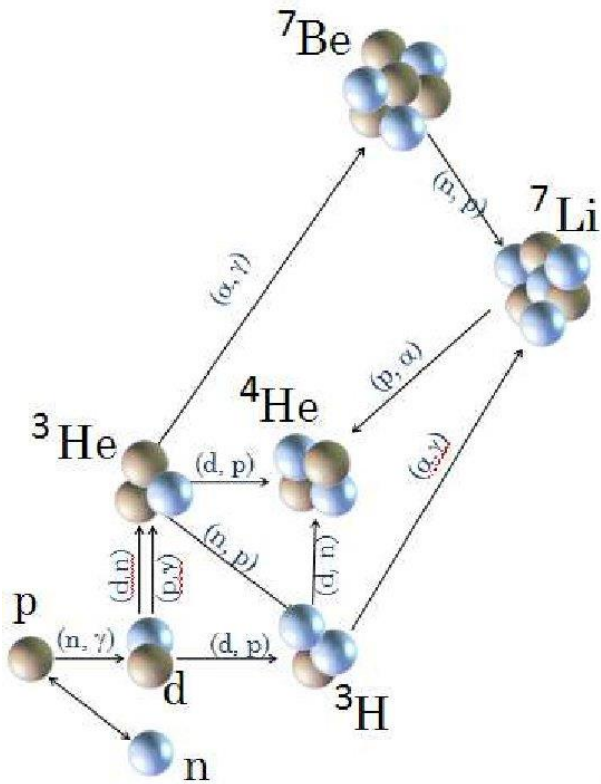
T-matrix

electromagnetic operator

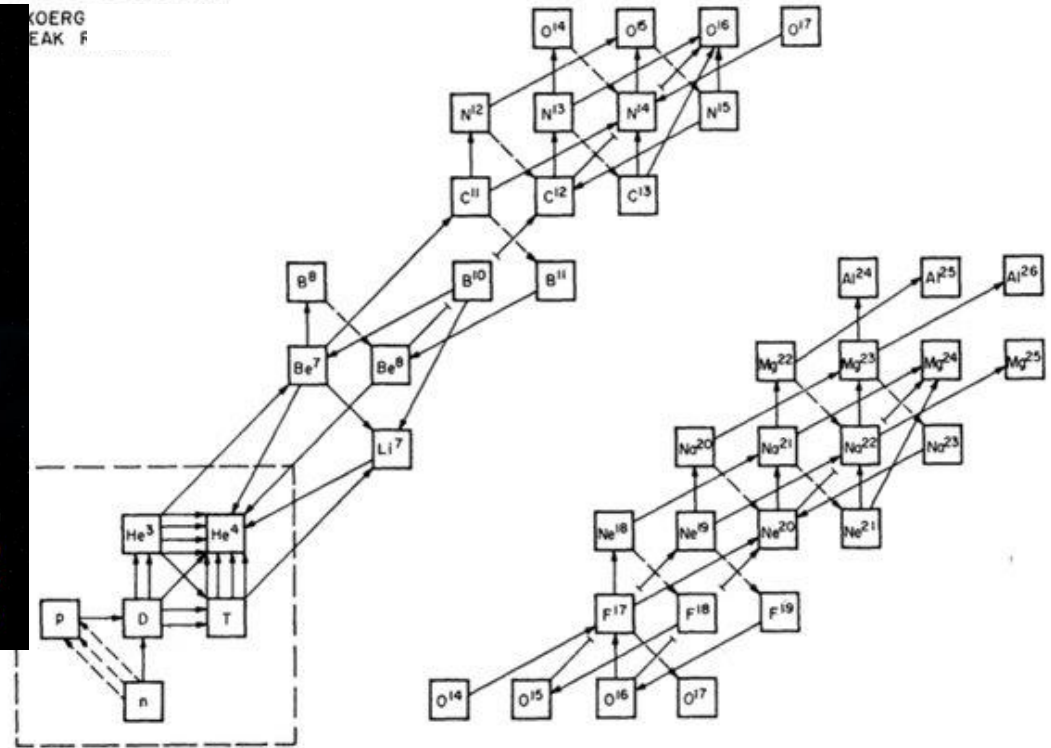
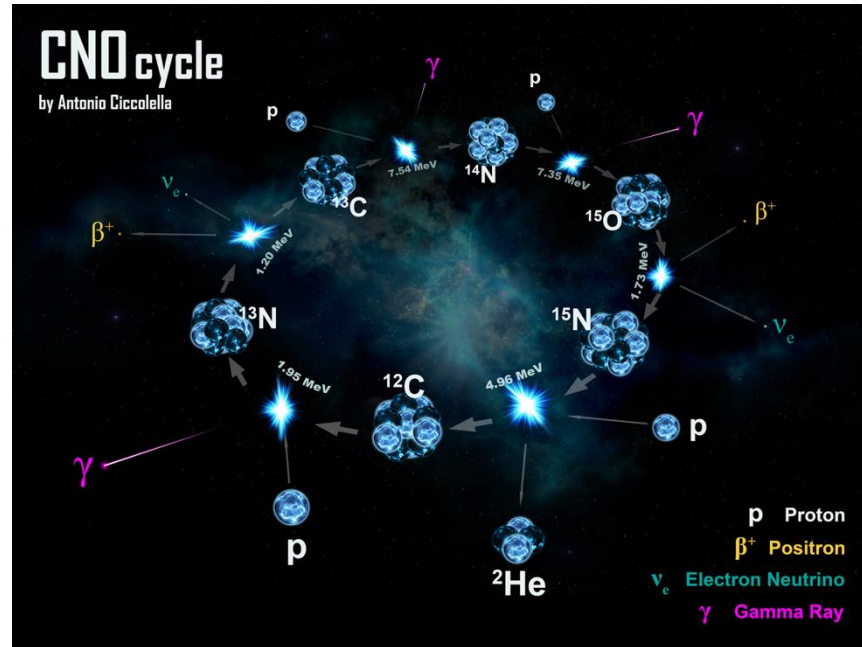


- solution of **6-body** Hamiltonian with chiral NN interaction
- No Core Shell Model with Continuum (**NCSMC**) combines **NCSM** with **scattering wf**
- only possible for **very light nuclei**

Nuclear reactions of astrophysical interest (light elements)

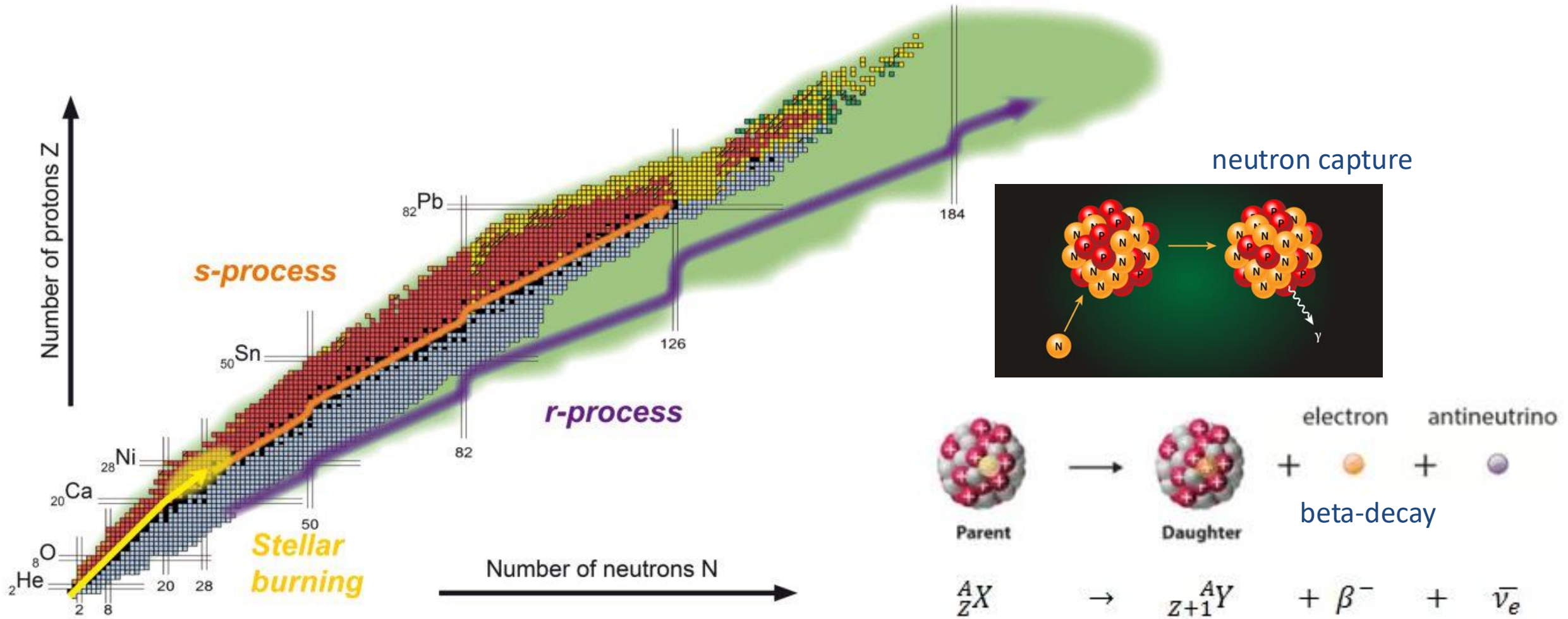


Big Bang nucleosynthesis

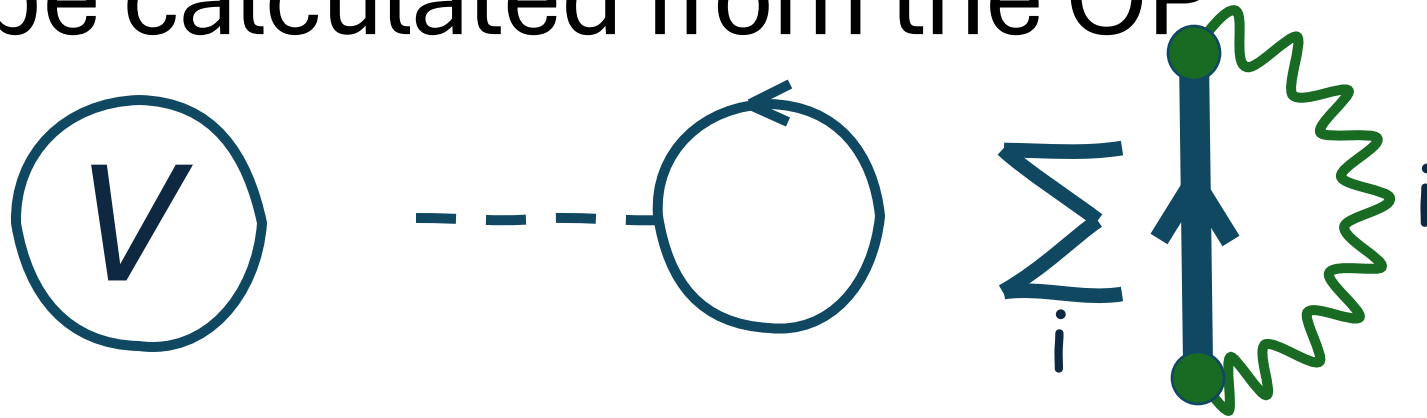


reaction networks

Nuclear reactions of astrophysical interest (heavy elements)



Both elastic and absorption cross sections can be calculated from the OP



$$(E - T - V(\mathbf{r}, \mathbf{r}'; E)) \phi = 0 \Rightarrow \text{elastic scattering from phase shifts}$$

$$\sigma_{\text{abs}} \sim \langle \phi | \text{Im}(\sum_i \text{polarization potential}) | \phi \rangle \Rightarrow \text{absorption from imaginary part of polarization potential}$$